

K-Field Spectrophotometer pic_id14C Analysis

Unwrapped CMB Trajectory, Multichannel Polarization, Rectilinear Plane-Stack Geometry, Reversals, and Nonlinear Channel Coupling

Field	Entry
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Acquisition interval	2026-08-04T10:55:12.483845+00:00 through 2026-08-04T22:53:19.473927+00:00
Generation epoch	1785888480
Generation UTC	2026-08-05T00:08:00Z
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This document is an archival technical reference. It presents the complete measured-data analysis, dimensional derivations, validation tests, negative results, and reproducibility ledger.

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Abstract

This archival reference develops a self-contained analysis of a twelve-hour rotating spectrophotometer record. The source contains 84,330 measurements, 937 complete 90-position revolutions, and 933 overlapping five-revolution profiles. Each profile is registered to the inertial position and orientation of its center revolution. The laboratory motion is unwrapped as the sum of the supplied CMB translation vector, Earth rotation, laboratory position, and the measured device rotation. This treatment preserves the fact that successive local circles are not closed curves in inertial space.

The channel state is low dimensional but not scalar. At periods of 12-64 revolutions, a broadband common mode carries 65.77% of standardized variance, while a second component carries 17.18% of total variance and 50.18% of the variance remaining after removal of the common mode. This second component separates 415-480 nm from 515-680 nm and NIR, establishing a spectral polarization mode.

The polarized mode contains a resolved chirp whose period contracts from 44.52 to 29.57 revolutions. The changing frequency is explained by traversal through a fixed stack of parallel planes. The fitted primitive spacing is 1,566.144 km, and the axial plane normal is RA 291.732 deg, Dec +35.972 deg. The normal is 89.9469 deg from the supplied CMB velocity, so the comparatively small Earth-rotation velocity projected onto the plane normal materially changes the crossing rate. The velocity-derived model explains 94.98% of the frequency track and retains approximately 60% phase-prediction power in held-out data.

The fitted phase divides the spectrum into nearly opposite states. The amplitude-weighted phase of 415-480 nm is 246.58 deg; the phase of 515-680 nm plus NIR is 70.82 deg; their separation is 175.76 deg. Twenty-four of twenty-nine fitted polarization reversals are matched within five revolutions. Independent device-angle harmonic analysis identifies five stable pairs: 415-NIR, 445-clear, 480-630, 515-680, and 555-590. Nonlinear screening further shows strong quadrature and curved dependencies that have near-zero Pearson correlation but high mutual information.

Core result. The twelve-hour record resolves one phase-coherent rectilinear plane-stack mode in the unwrapped CMB-plus-laboratory trajectory. The mode has a measured primitive spacing, a stable inertial normal, a velocity-governed chirp, a nearly antipodal spectral polarization, and repeatable phase reversals. The record does not yet establish two additional independent plane families and therefore does not define a complete three-dimensional lattice cell.

1. Experimental Record and Measurement Geometry

1.1 Source record

Quantity	Value
Source file	pic_id_14C.csv
SHA-256	93835d3dea837ab9f7c5a43e73715e2f23c2f6874b6488db667fcd1025e1a273
Measurements	84,330
Complete revolutions	937
Five-turn profiles	933
First effective UTC	2026-08-04T10:55:12.483845+00:00
Last effective UTC	2026-08-04T22:53:19.473927+00:00
Duration	11.968608356 h
Earth rotation	180.020659877 deg
Median revolution period	45.985105 s
CMB speed	930825.217271 m/s
CMB translation distance	40106456904.610 m
Device radius	0.041529 m
Lab location	lat 42.129 deg, lon -83.144 deg, altitude 213.36 m

The experiment covers 11 h 58 min 7.0 s and almost exactly 180 deg of Earth rotation. That duration is important because the laboratory vertical and north directions sweep through approximately one inertial half-turn. The detector therefore samples a large family of inertial directions while the laboratory simultaneously translates approximately 4.01×10^{10} m along the supplied CMB vector.

1.2 Device rotation and five-revolution overlap

$$S_{c,k}(\theta) = \sum_{j=0..4} x_{c,k+j}(\theta)$$

For channel c , revolution k , and device angle θ , five consecutive revolutions are added element by element. The stack advances by one revolution, so neighboring profiles share four revolutions. Summation preserves channel counts and improves the signal-to-noise ratio of features that require several revolutions to resolve. The center revolution supplies the time and inertial registration. The maximum Earth-rotation displacement between the center and either edge of a five-turn stack is approximately 0.388 deg.

The device direction is defined by index zero pointing down, increasing index sweeping toward north, then up, south, and down. With $\theta = 4 \text{ deg} \times \text{my_index}$, the inertial direction is

$$u(t, \theta) = \sin(\theta) \text{ north_ICRS}(t) - \cos(\theta) \text{ up_ICRS}(t)$$

ICRS geometry: device scan, CMB direction, and resolved plane normal

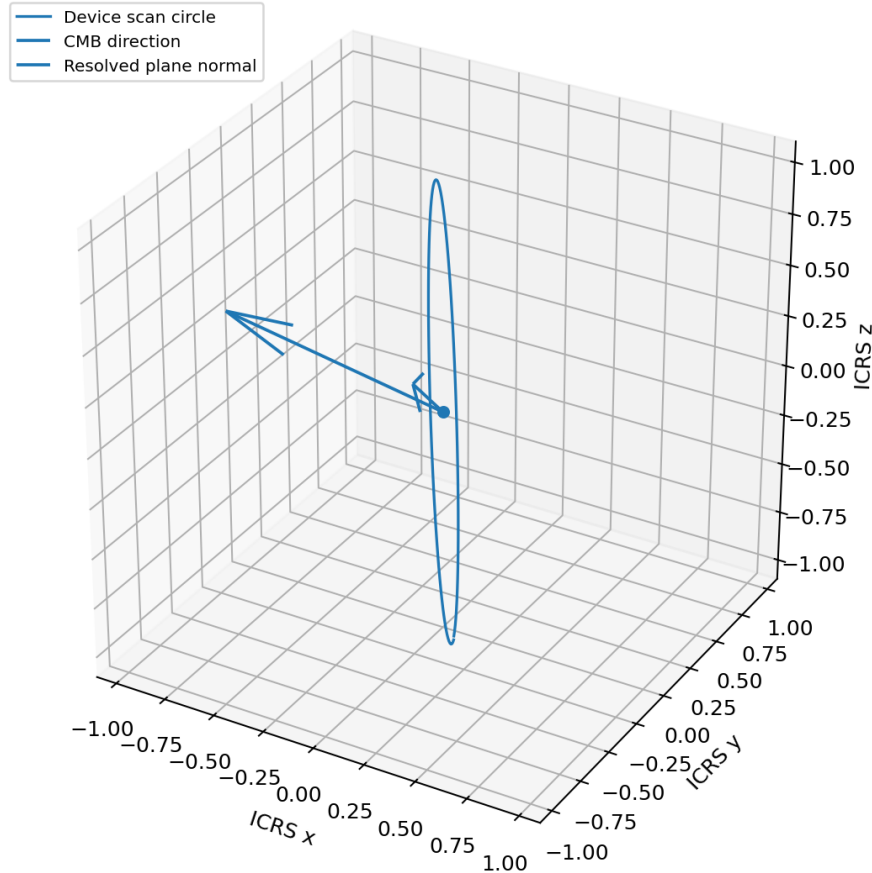


Figure 1. ICRS unit-vector geometry at the middle of the record. The complete measured device scan circle is shown with the supplied CMB direction and the resolved plane normal. All vectors are dimensionless unit directions.

1.3 Data-health ledger

The record contains 84,183 rows with `health_flag = 1` and 147 rows with `health_flag = 0`. The analysis retains all rows to preserve continuity and reproduce the established pipeline. The flagged rows remain within the channel-value ranges occupied by healthy measurements. This is a retention decision, not an assertion that the flag is irrelevant; the machine-readable companion preserves the counts and source hash.

2. Unwrapped CMB-Plus-Laboratory Trajectory

2.1 Translation and modulation equation

$$R(t, \theta) = g(t - t_0) + r_{\text{lab_ICRS}}(t) + r_d u_{\text{ICRS}}(t, \theta)$$

The first term is linear translation along the supplied CMB velocity g . The second term is the inertial position of the laboratory as Earth rotates. The third term is the detector offset from the east-west rotation axis. Coordinates are reported relative to the first five-turn center sample; this removes a physically irrelevant absolute origin without closing the trajectory.

Trajectory quantity	Value
CMB velocity x	-880221.981635 m/s
CMB velocity y	-143911.436562 m/s
CMB velocity z	266335.026949 m/s
CMB unit vector	-0.945636157362, -0.154606293310, 0.286127859460
Basis e1	0.161352195205, -0.986896888790, 0.000000000000
Basis e2	0.282378694297, 0.046167358233, 0.958191446445
Along-track span	39943528375.930 m
Transverse-1 span	6666959.767 m
Transverse-2 span	2586276.940 m

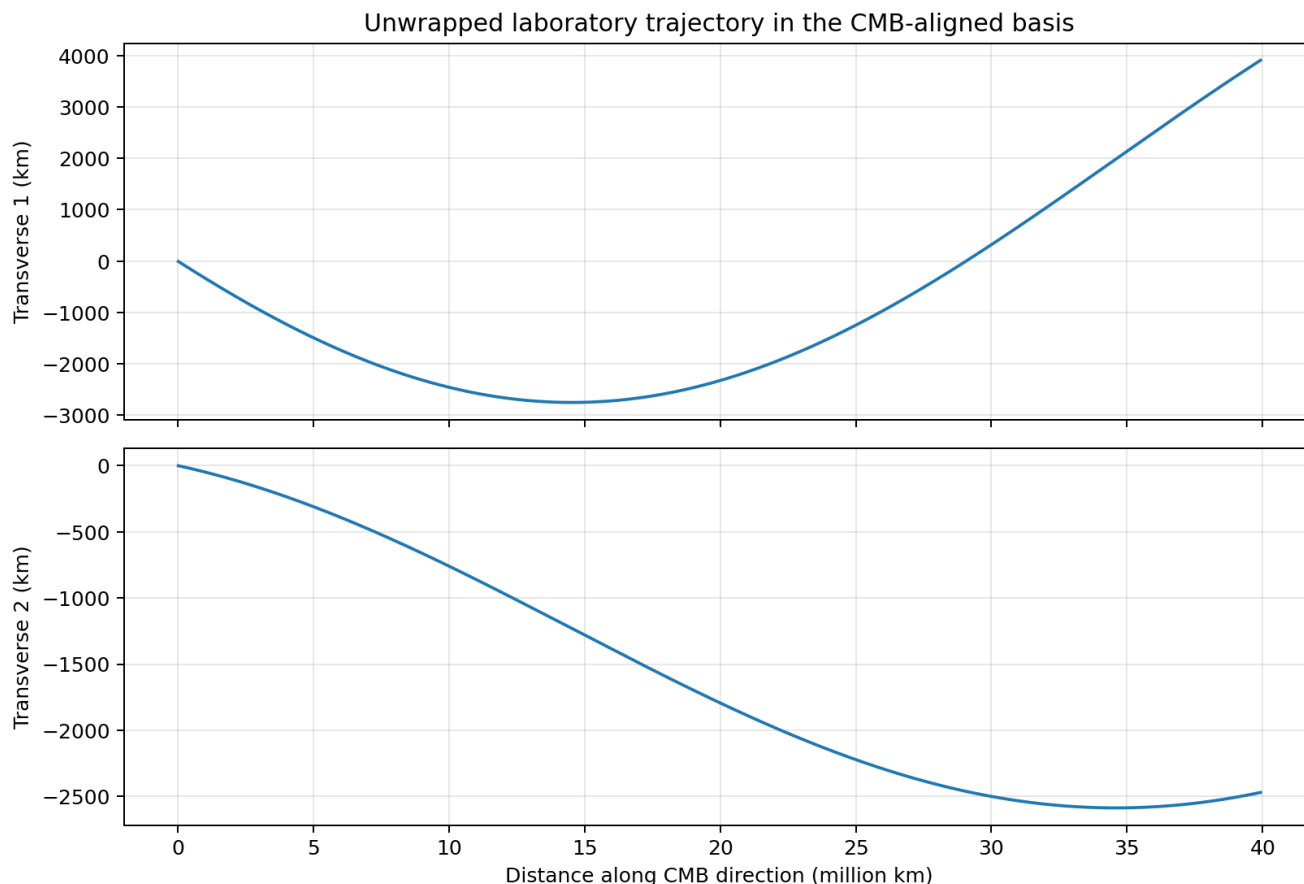


Figure 2. The unwrapped laboratory trajectory in a basis aligned with the supplied CMB vector. The horizontal axis is actual distance traveled along the CMB direction. The transverse excursions are generated mainly by Earth rotation and laboratory motion.

The along-track displacement is approximately forty million kilometres, while Earth rotation produces transverse excursions of 6,667 km and 2,586 km in the chosen CMB-orthogonal basis. The large scale separation is central to the plane-stack result: a plane normal nearly perpendicular to the CMB direction converts a small projected component of the enormous translation into a crossing speed that is substantially modulated by Earth rotation.

3. Multiscale Channel-State Reconstruction

3.1 Period bands

The logarithmic five-turn DC signal of every channel is separated into three period bands. The 3-12-turn band captures rapidly resolved rotational structure. The 12-64-turn band contains the principal polarized chirp. The 64-300-turn band captures slower channel relationships and nonlinear loops. The decomposition is designed to preserve the known 3-10-turn signal scale rather than remove it as a trend.

Band	PC1 variance	PC1+PC2	PC1+PC2+PC3	Participation dimension
3-12 turns	78.25%	84.58%	89.17%	1.611

Band	PC1 variance	PC1+PC2	PC1+PC2+PC3	Participation dimension
12-64 turns	65.77%	82.95%	88.62%	2.137
64-300 turns	55.19%	89.99%	94.15%	2.332

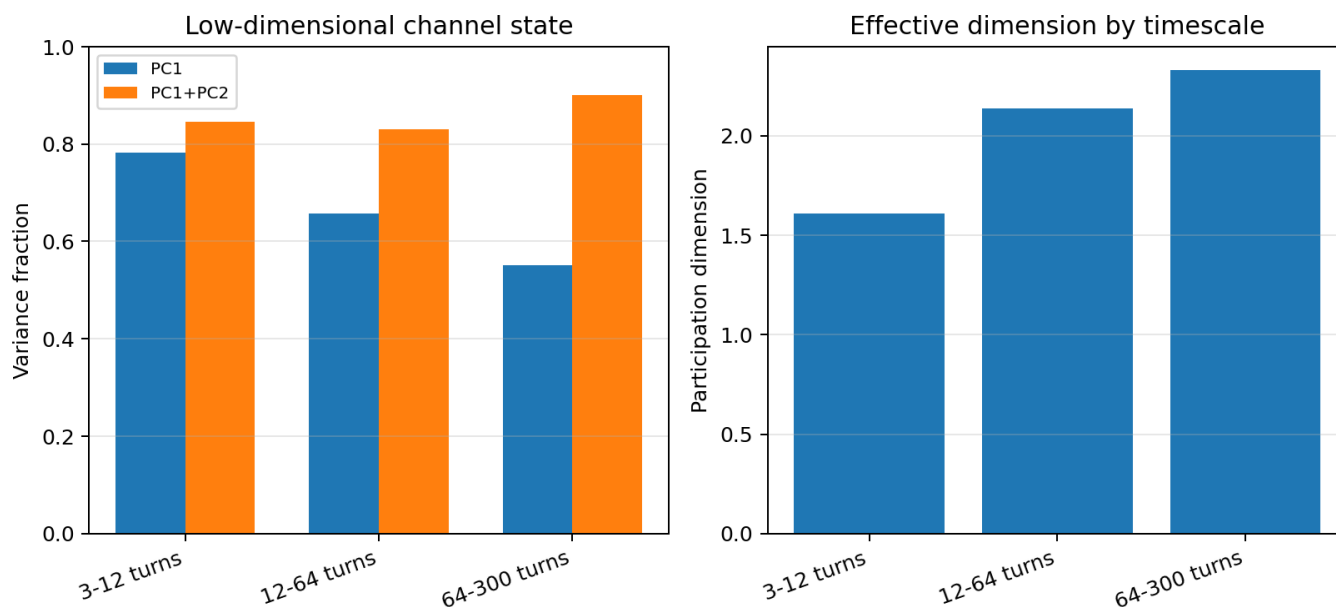


Figure 3. Variance concentration and participation dimension across the three period bands. The channel state becomes more two-dimensional at longer periods.

3.2 Interpretation of effective dimension

At 3-12 turns, PC1 alone carries 78.25% of standardized variance and the participation dimension is 1.61. The response is therefore dominated by a common broadband motion with smaller channel-specific structure. At 12-64 turns, PC1 falls to 65.77%, while PC1 and PC2 together carry 82.95%; the effective dimension rises to 2.14. At 64-300 turns, PC1 carries 55.19% and PC1+PC2 carries 90.00%, with dimension 2.33. The system is thus well represented as a common mode plus a polarized mode, not as ten independent signals.

4. Spectral Polarization Mode

4.1 Mid-band principal components

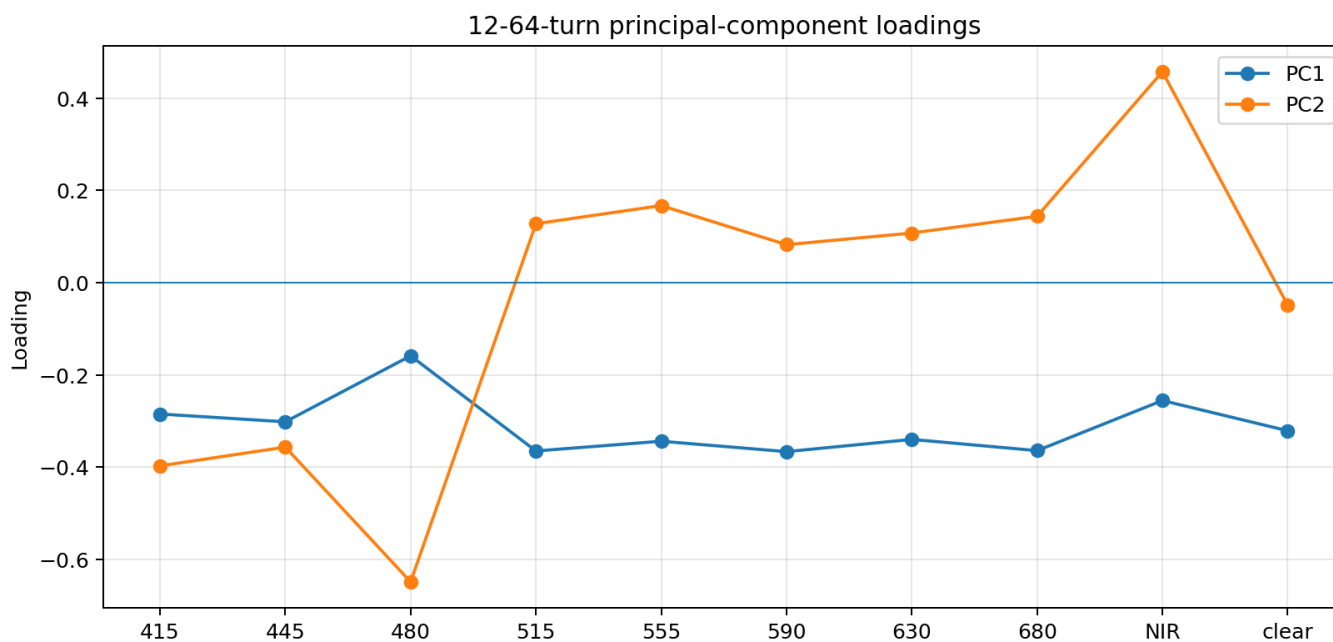


Figure 4. Loadings of the first two principal components in the 12-64-turn band. PC1 is broadband. PC2 changes sign between the blue channels and the lower-energy plus NIR channels.

Channel	PC2 loading
415	-0.397270
445	-0.356598
480	-0.648711
515	+0.127430
555	+0.166822
590	+0.082090
630	+0.107312
680	+0.143670
NIR	+0.456962
clear	-0.049094

PC2 carries 17.18% of total standardized mid-band variance and 50.18% of all variance left after the common mode. The positive pole is 415, 445, and 480 nm. The negative pole is 515, 555, 590, 630, 680, and NIR. Clear is near the center of this polarization.

4.2 Phase-defined polarization

The amplitude-weighted blue-channel phase is 246.577 deg with resultant length 0.993813. The 515-680 nm plus NIR phase is 70.817 deg with resultant length 0.959306. Their axial separation is 175.759 deg. Resultant lengths close to one mean that the channels within each pole share a sharply defined phase.

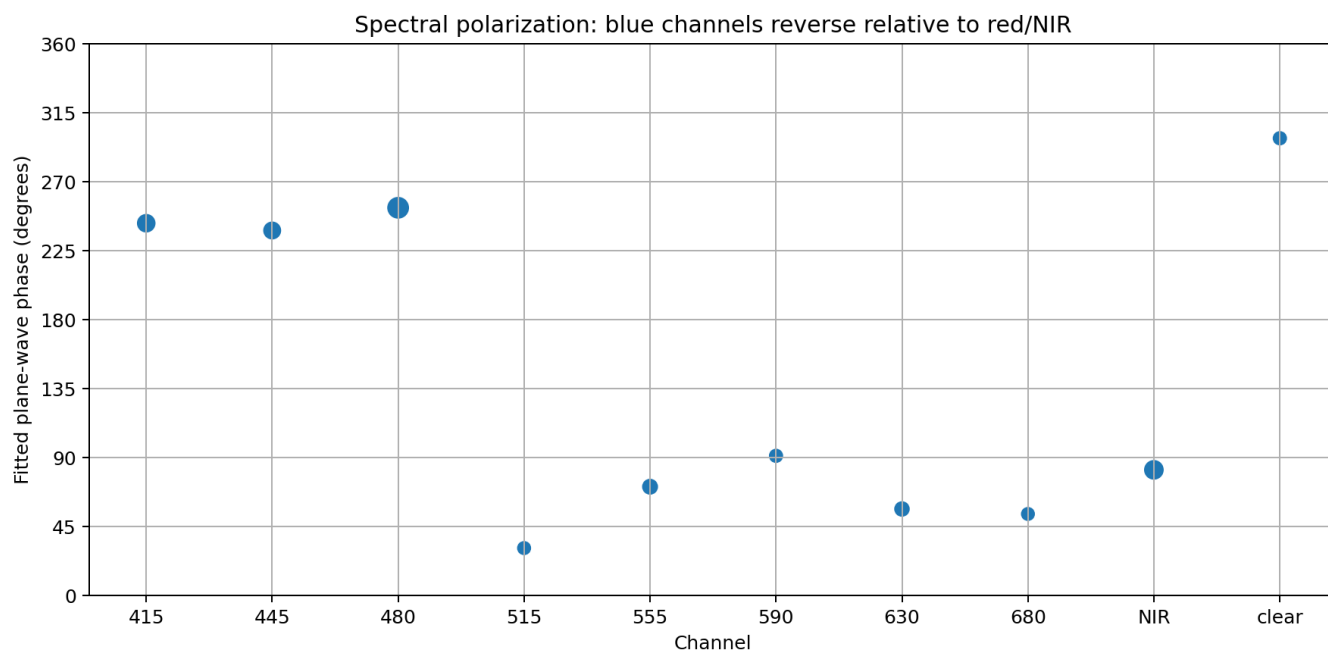


Figure 5. Channel phases relative to the resolved plane-wave coordinate. The blue channels occupy one phase group; lower-energy visible channels and NIR occupy the nearly antipodal group.

5. Rectilinear Plane-Stack Model

5.1 From changing period to plane geometry

For a stack of parallel planes with wavevector q in cycles per metre, the unwrapped spatial phase is

$$\Phi(t) = 2 \pi q \cdot R(t) + \phi_{i0}$$

The instantaneous crossing frequency is the directional derivative of phase along the trajectory:

$$f(t) = q \cdot v(t)$$

The measured chirp therefore supplies a direct geometric constraint. A fixed q must project the changing CMB-plus-laboratory velocity into the observed crossing frequency. Unlike an arbitrary time polynomial, this equation has a physical direction, a physical spacing, and a specified response to Earth rotation.

Plane quantity	Measured value
Wavevector q_x	1.913377581942e-07 cycles/m
Wavevector q_y	-4.800188656215e-07 cycles/m
Wavevector q_z	3.750569111340e-07 cycles/m
Plane-normal x	0.299662563394
Plane-normal y	-0.751778870555
Plane-normal z	0.587393290639
Plane-normal RA	291.732456586 deg
Plane-normal Dec	+35.972245579 deg
Primitive spacing d	1566144.425556 m
Reversal-sheet spacing $d/2$	783072.212778 m
Angle to CMB axis	89.946853887 deg
CMB velocity projection	863.409758 m/s
Earth-rotation projection range	-79.288143 to 270.981155 m/s
Total normal-speed range	784.121614 to 1134.390913 m/s

5.2 Resolved primitive cell and plane nodes

Resolved one-dimensional primitive cell and polarization phase landmarks

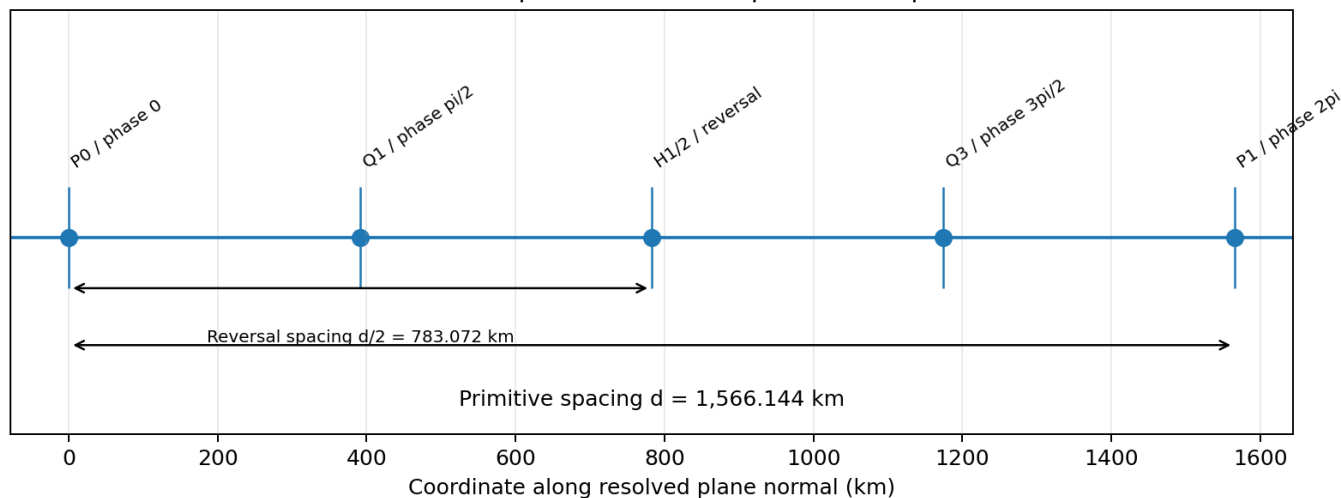


Figure 6. Dimensionally correct one-dimensional primitive cell along the resolved plane normal. P0 and P1 are equal-phase planes separated by $d = 1,566.144$ km. H1/2 is the polarization-reversal sheet at $d/2 = 783.072$ km. Q1 and Q3 are quarter-cycle phase landmarks.

Node	Cycle fraction	Normal coordinate (m)	Normal coordinate (km)	Phase
P0	0.00	0.000	0.000	0
Q1	0.25	391536.106	391.536	$\pi/2$
H1/2	0.50	783072.213	783.072	π
Q3	0.75	1174608.319	1174.608	$3\pi/2$
P1	1.00	1566144.426	1566.144	2π

This is the complete primitive geometry resolved by the present record. The data determine translation along one normal direction and phase repetition after one spacing d . They do not determine two independent tangential periods. Consequently, no dimensionally justified three-dimensional parallelepiped or full lattice cell can be drawn. Any such volumetric cell would require at least two additional independent plane families with successful phase closure.

5.3 Why Earth rotation produces the chirp

The fitted normal is almost perpendicular to the CMB direction. The full CMB speed is approximately 930.8 km/s, but only 863.4 m/s projects onto the normal. Earth rotation adds between -79.3 and +271.0 m/s. This comparatively modest laboratory term changes the total normal speed from 784.1 to 1,134.4 m/s, a 44.7% increase. Because crossing period is spacing divided by normal speed, the predicted period contracts from 43.43 to 30.02 revolutions.

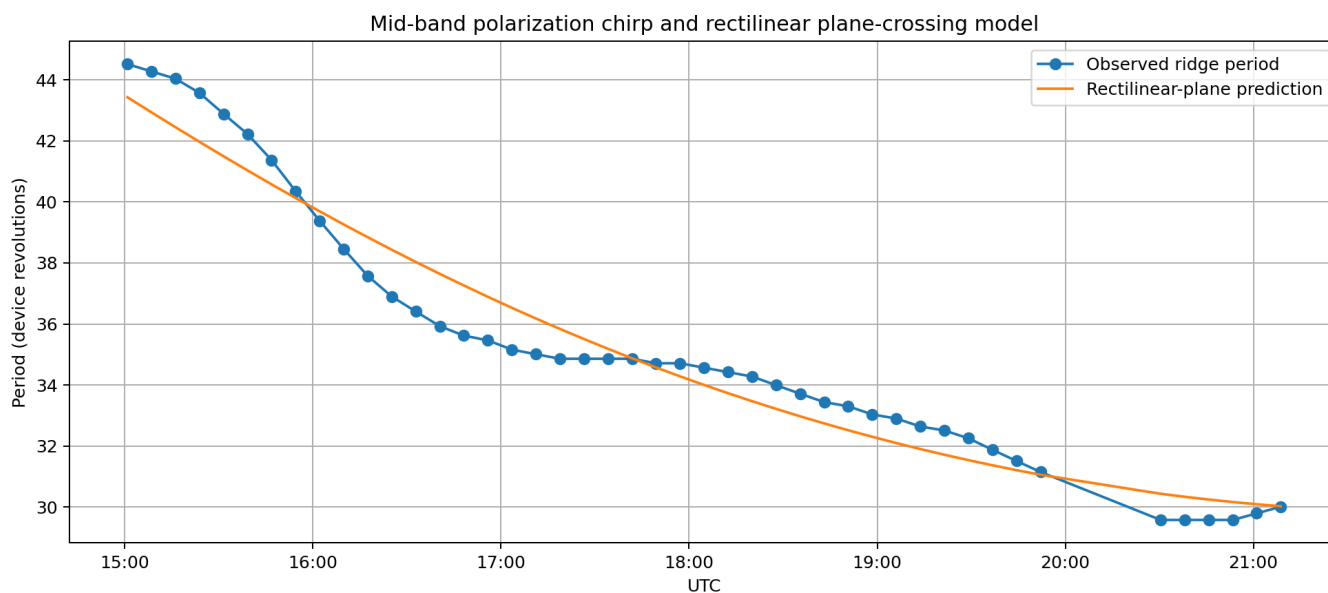


Figure 7. Observed chirp ridge and velocity-derived plane-crossing prediction. The observed period contracts from approximately 44.52 to 29.57 revolutions; the fixed-plane model predicts 43.43 to 30.02 revolutions.

6. Model Validation and Predictive Tests

6.1 Frequency and spatial-phase closure

Validation test	Result
Observed frequency vs. plane-normal velocity R^2	0.949759
PC2 vs. spatial phase R^2 , stacks 300-800	0.658970
Blue-minus-red/NIR contrast vs. spatial phase R^2	0.631155
Train stacks 300-550; predict 551-800	0.601514
Matched reversals	24 of 29
Median absolute reversal residual	1.968087 turns = 90.503 s

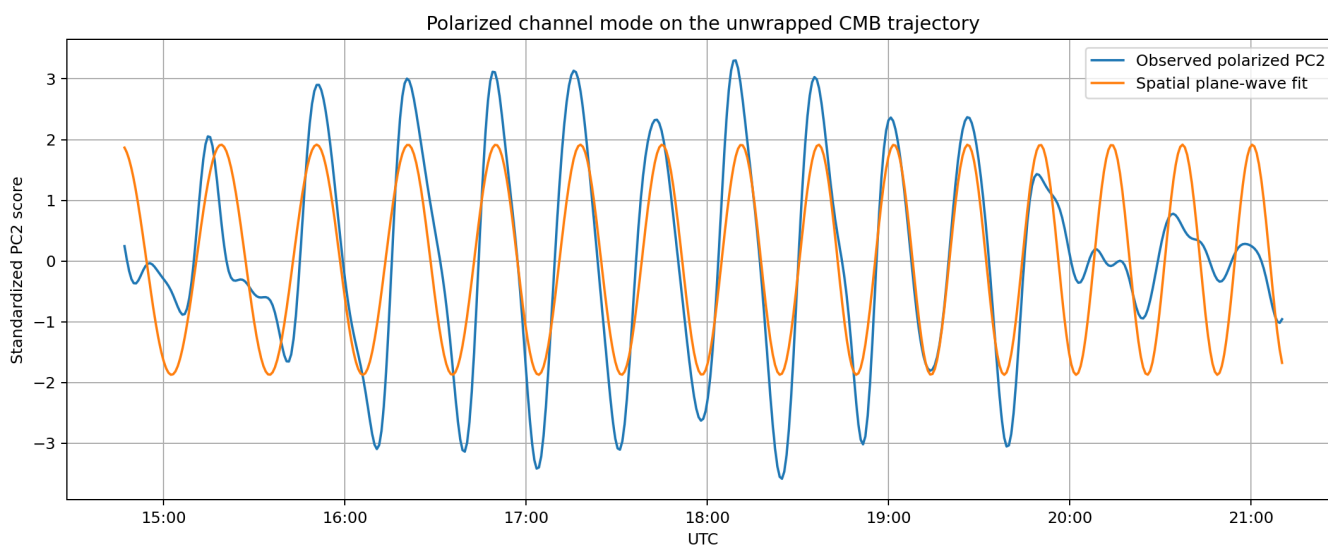


Figure 8. Mid-band PC2 and its fitted spatial plane wave through the resolved interval. The plot uses the unwrapped trajectory rather than a closed local angle.

6.2 Held-out comparison with generic time chirps

Model/test	Held-out result
Geometry model phase holdout, split A	0.600218
Geometry model phase holdout, split B	0.602986
Even/odd ridge frequency R^2 , split A	0.947013
Even/odd ridge frequency R^2 , split B	0.951700
Quadratic time-only chirp holdout R^2	-1.294589
Cubic time-only chirp holdout R^2	0.192865

A flexible time polynomial can imitate the observed chirp within the fitted interval. The decisive test is extrapolation. The fixed spatial geometry retains approximately 0.60 R^2 in the late interval, while the quadratic time-only chirp falls to -1.295 and the cubic chirp reaches only 0.193. The velocity-projection model therefore carries predictive structure that is absent from the generic time fits.

6.3 Sensitivity to ridge-window choices

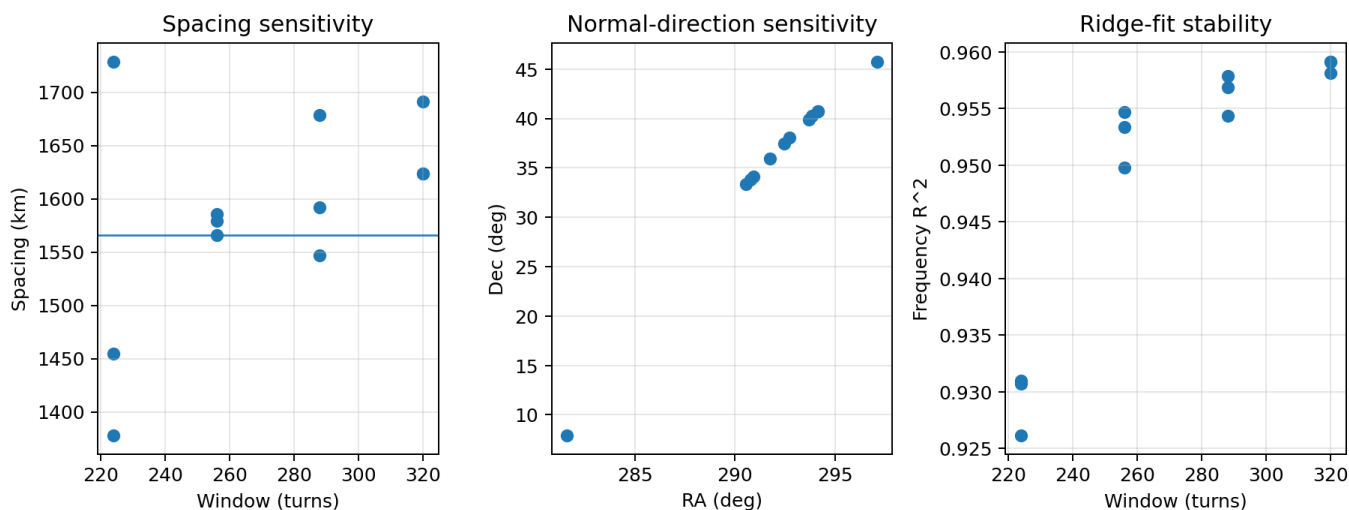


Figure 9. Sensitivity of plane spacing, normal direction, and frequency-fit quality to ridge-window and search-range choices. The central solution is surrounded by a stable family of high- R^2 fits.

Window	Period search	Ridge points	R^2	Spacing km	RA deg	Dec deg	CMB angle deg
224	18-60	42	0.930965	1454.445	293.688	+39.926	89.950317
224	20-64	41	0.926163	1378.131	297.081	+45.782	89.951905
224	22-70	40	0.930728	1728.910	281.613	+7.936	89.945270
256	18-60	47	0.954702	1578.924	290.903	+34.149	89.946729
256	20-64	45	0.949759	1566.144	291.732	+35.972	89.946854
256	22-70	46	0.953348	1585.501	290.770	+33.850	89.946546
288	18-60	48	0.957861	1546.971	293.846	+40.236	89.946605
288	20-64	47	0.956872	1591.784	292.724	+38.043	89.945372
288	22-70	45	0.954348	1678.373	290.536	+33.326	89.943052
320	18-60	48	0.959153	1623.699	294.117	+40.754	89.943477
320	20-64	48	0.959153	1623.699	294.117	+40.754	89.943477
320	22-70	47	0.958126	1691.418	292.449	+37.492	89.941588

7. Polarization Reversals and Plane Crossings

7.1 Reversal mechanism

A sinusoidal plane response changes sign every half cycle. Equivalent-phase planes are separated by d , while sign-reversal sheets are separated by $d/2$. The measured blue-minus-red/NIR contrast is therefore expected to reverse as the unwrapped trajectory crosses successive half-cycle sheets.

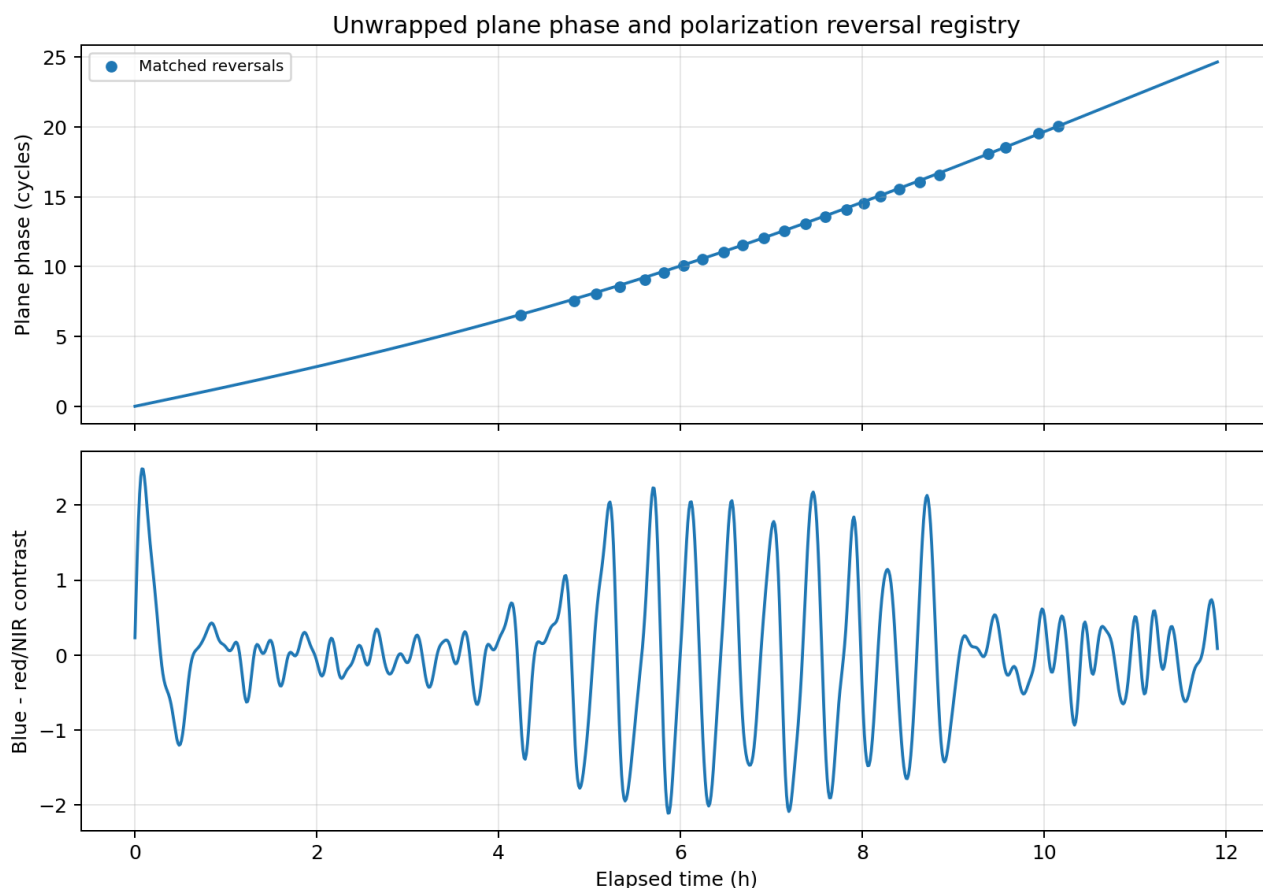


Figure 10. Unwrapped plane phase and blue-minus-red/NIR contrast. Matched measured reversals are placed on the fitted phase track.

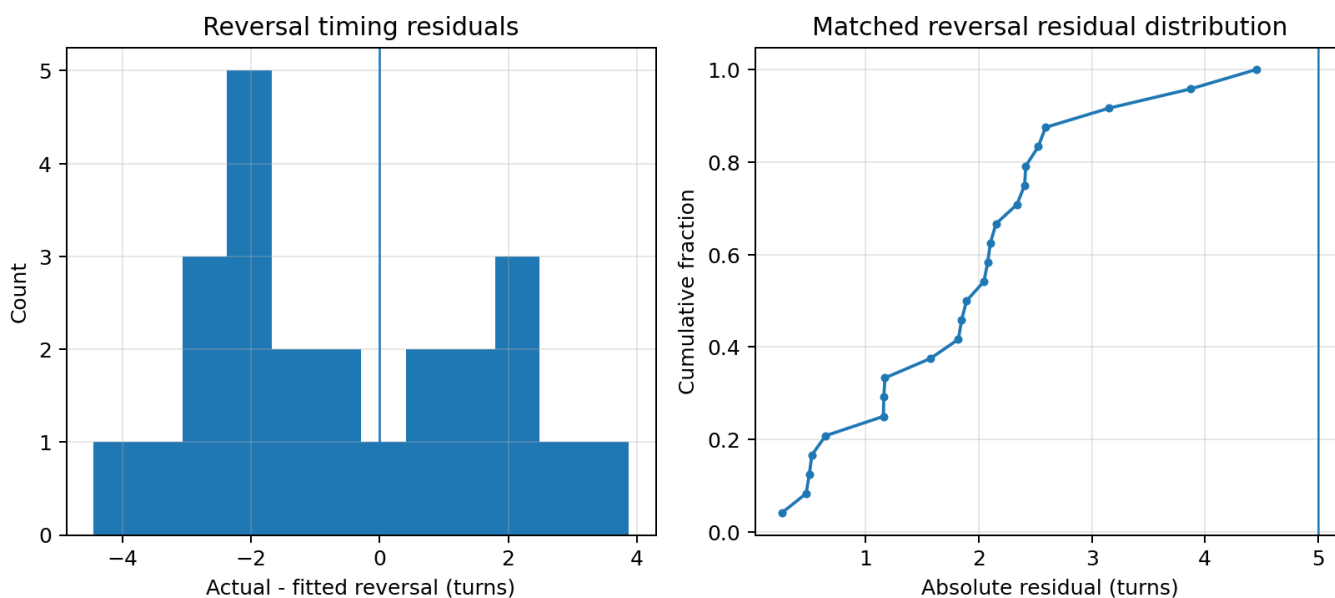


Figure 11. Timing residuals for the 24 matched reversals. The median absolute residual is 1.97 revolutions, or approximately 90.5 s.

7.2 Complete reversal-event ledger

Event	Fitted stack	Fitted UTC	Actual stack	Residual turns	Residual s	Plane cycle	Normal km	Matched
0	309.865	2026-08-04T14:54:36.378956032+00:00				6.063172	9495.803	no
1	331.392	2026-08-04T15:11:06.284646400+00:00	329.240	-2.152	-99.0	6.560296	10274.371	yes
2	352.643	2026-08-04T15:27:23.481088768+00:00				7.063175	11061.952	no
3	373.159	2026-08-04T15:43:06.932182272+00:00	375.685	+2.526	+116.2	7.560302	11840.525	yes
4	393.446	2026-08-04T15:58:39.801087232+00:00	394.605	+1.159	+53.3	8.063184	12628.111	yes
5	413.066	2026-08-04T16:13:42.009861376+00:00	415.111	+2.045	+94.0	8.560300	13406.667	yes
6	432.501	2026-08-04T16:28:35.729156096+00:00	436.377	+3.876	+178.2	9.063182	14194.252	yes
7	451.332	2026-08-04T16:43:01.671272448+00:00	452.902	+1.570	+72.2	9.560300	14972.811	yes
8	470.021	2026-08-04T16:57:21.060753408+00:00	469.548	-0.472	-21.7	10.063180	15760.393	yes
9	488.162	2026-08-04T17:11:15.282444032+00:00	486.059	-2.102	-96.7	10.560305	16538.963	yes
10	506.199	2026-08-04T17:25:04.711097856+00:00	504.380	-1.819	-83.6	11.063195	17326.561	yes
11	523.738	2026-08-04T17:38:31.225772032+00:00	520.584	-3.153	-145.0	11.560277	18105.063	yes
12	541.209	2026-08-04T17:51:54.647554304+00:00	538.804	-2.405	-110.6	12.063197	18892.708	yes
13	558.227	2026-08-04T18:04:57.229809920+00:00	556.380	-1.847	-85.0	12.560307	19671.255	yes
14	575.207	2026-08-04T18:17:58.047281408+00:00	574.707	-0.500	-23.0	13.063198	20458.855	yes
15	591.774	2026-08-04T18:30:39.867270144+00:00	591.516	-0.258	-11.9	13.560274	21237.347	yes
16	608.333	2026-08-04T18:43:21.305939456+00:00	610.224	+1.891	+87.0	14.063197	22024.998	yes
17	624.514	2026-08-04T18:55:45.370856448+00:00	625.037	+0.523	+24.1	14.560287	22803.513	yes
18	640.709	2026-08-04T19:08:10.126890240+00:00	639.540	-1.169	-53.8	15.063166	23591.093	yes
19	656.562	2026-08-04T19:20:19.092834304+00:00	655.406	-1.155	-53.1	15.560282	24369.649	yes
20	672.452	2026-08-04T19:32:29.816078080+00:00	673.096	+0.644	+29.6	16.063189	25157.275	yes
21	688.027	2026-08-04T19:44:26.034112+00:00	690.106	+2.078	+95.6	16.560300	25935.822	yes
22	703.661	2026-08-04T19:56:24.941069312+00:00				17.063168	26723.385	no
23	719.007	2026-08-04T20:08:10.637355520+00:00				17.560297	27501.961	no
24	734.432	2026-08-04T20:19:59.959646976+00:00	732.017	-2.416	-111.1	18.063192	28289.568	yes
25	749.592	2026-08-04T20:31:37.076019456+00:00	747.002	-2.590	-119.1	18.560278	29068.075	yes
26	764.851	2026-08-04T20:43:18.744523264+00:00				19.063160	29855.662	no
27	779.868	2026-08-04T20:54:49.283744512+00:00	775.414	-4.454	-204.8	19.560272	30634.210	yes
28	795.002	2026-08-04T21:06:25.216261376+00:00	792.664	-2.337	-107.5	20.063178	31421.834	yes

8. Channel-Resolved Plane-Wave Responses

Each common-mode-removed channel is fitted to the same spatial phase. The fitted phase identifies spectral polarity; R^2 measures how much of that channel component is explained by the plane wave; the circular-shift percentile compares the actual ICRS registry with autocorrelation-preserving shifts.

Channel	Plane R^2	Amplitude	Phase deg	Null percentile	Null mean R^2	Null max R^2
415	0.372062	0.740841	242.975	100.000%	0.049515	0.111326
445	0.457417	0.652038	238.230	100.000%	0.021626	0.106126
480	0.601312	1.251609	253.054	100.000%	0.027890	0.071168
515	0.123141	0.173917	30.955	97.656%	0.030918	0.141052
555	0.230059	0.387285	71.003	100.000%	0.050199	0.131324
590	0.138933	0.193531	91.164	98.438%	0.042442	0.148205
630	0.206067	0.335646	56.471	98.438%	0.047151	0.217830
680	0.107285	0.163760	53.226	88.281%	0.043687	0.157268
NIR	0.540354	0.904296	82.014	100.000%	0.033795	0.230617
clear	0.059378	0.197769	298.433	67.969%	0.042714	0.140660

The strongest plane responses are 480 nm, NIR, 445 nm, and 415 nm. The blue channels cluster near 238-253 deg. The lower-energy visible channels and NIR cluster near 31-91 deg. Clear has weak registry and does not define either principal pole.

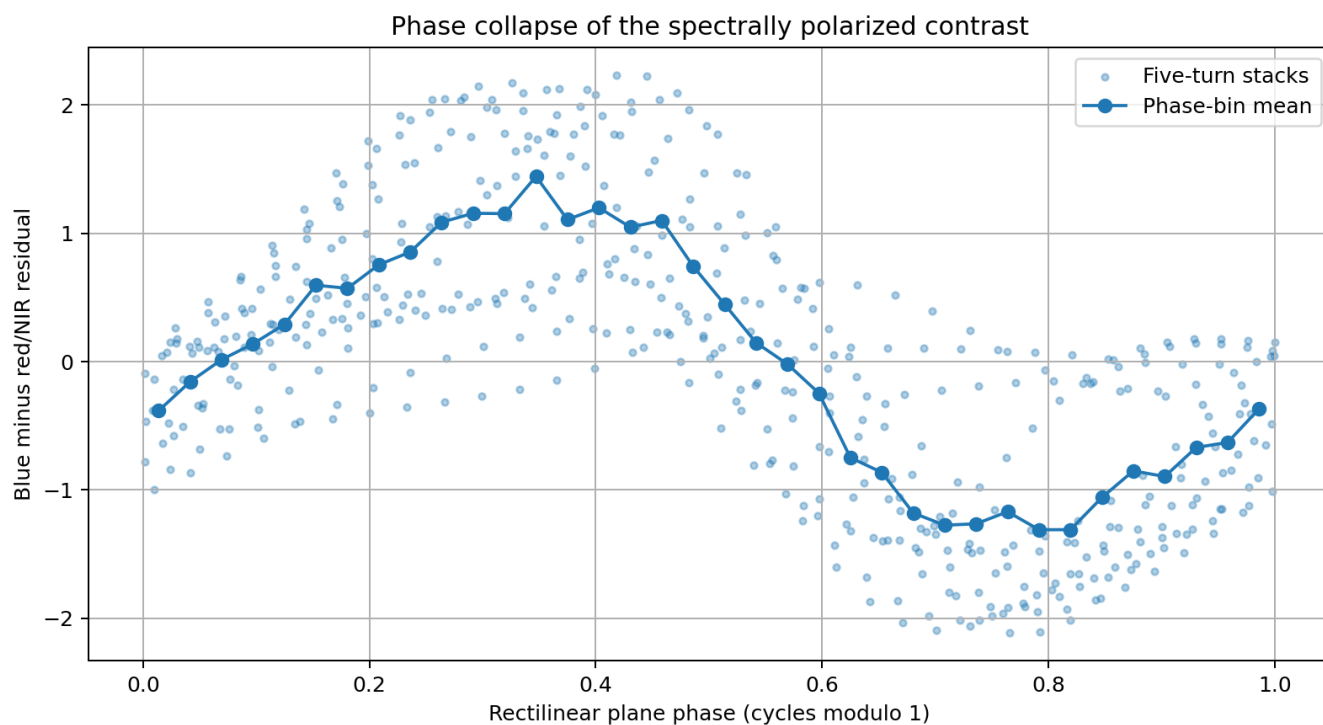


Figure 12. Blue-minus-red/NIR contrast folded by the fitted spatial phase. The coherent periodic modulation appears after mapping the time series into the unwrapped plane coordinate.

9. Hidden Device-Angle Channel Pairing

The plane-wave polarization describes slow evolution along the unwrapped trajectory. A separate analysis decomposes each five-turn angular profile into device harmonics. Complex coherence then tests whether two channels share the same angular waveform and phase. Maximum-weight matching produces the same five pairs for the first and third harmonics.

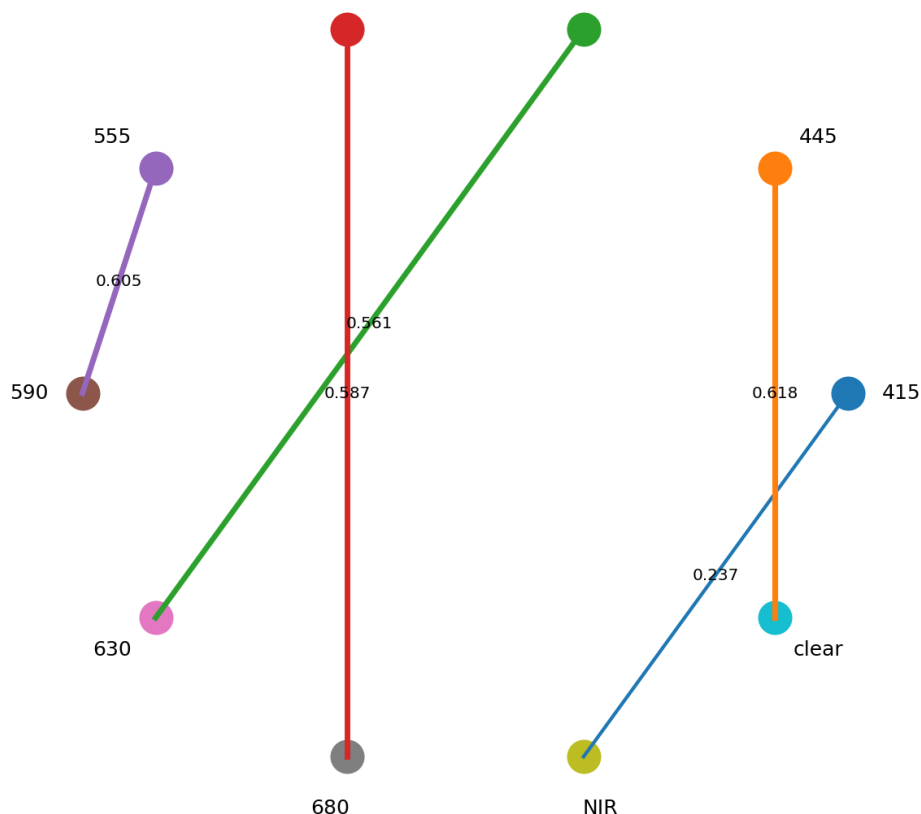
Matched channel pairs from device-angle harmonics $m=1$ through $m=3$ 

Figure 13. Stable channel-pair graph obtained from device-angle harmonics $m=1$ through $m=3$. Edge labels are the mean coherence over the first three harmonics.

Pair	m1 coh.	m1 phase	m2 coh.	m2 phase	m3 coh.	m3 phase	Mean coh.
415-NIR	0.372950	+0.544	0.193852	-2.952	0.143373	+5.066	0.236725
445-clear	0.701264	+0.884	0.640535	-7.290	0.511577	-5.981	0.617792
480-630	0.724214	+2.291	0.520604	-3.968	0.437841	-2.911	0.560886
515-680	0.711428	+0.700	0.581389	-3.669	0.468386	-7.251	0.587068
555-590	0.763762	-1.131	0.569354	+4.197	0.480899	-2.652	0.604671

The angular-profile grammar is 415-NIR, 445-clear, 480-630, 515-680, and 555-590. First-harmonic phase offsets are all within 2.3 deg of zero. The first-harmonic coherences exceed every one of the 128 circular-shift nulls. These synchronous angular pairs coexist with the slower blue-versus-red/NIR temporal polarization; they are different projections of the coupled channel state.

10. Nonlinear Channel Dependencies

10.1 Why linear correlation is insufficient

Pearson correlation detects a straight-line relation. A phase loop, curved relation, or state-dependent coupling can have near-zero Pearson correlation even when one channel strongly constrains another. Mutual information and distance correlation therefore complement complex phase coherence.

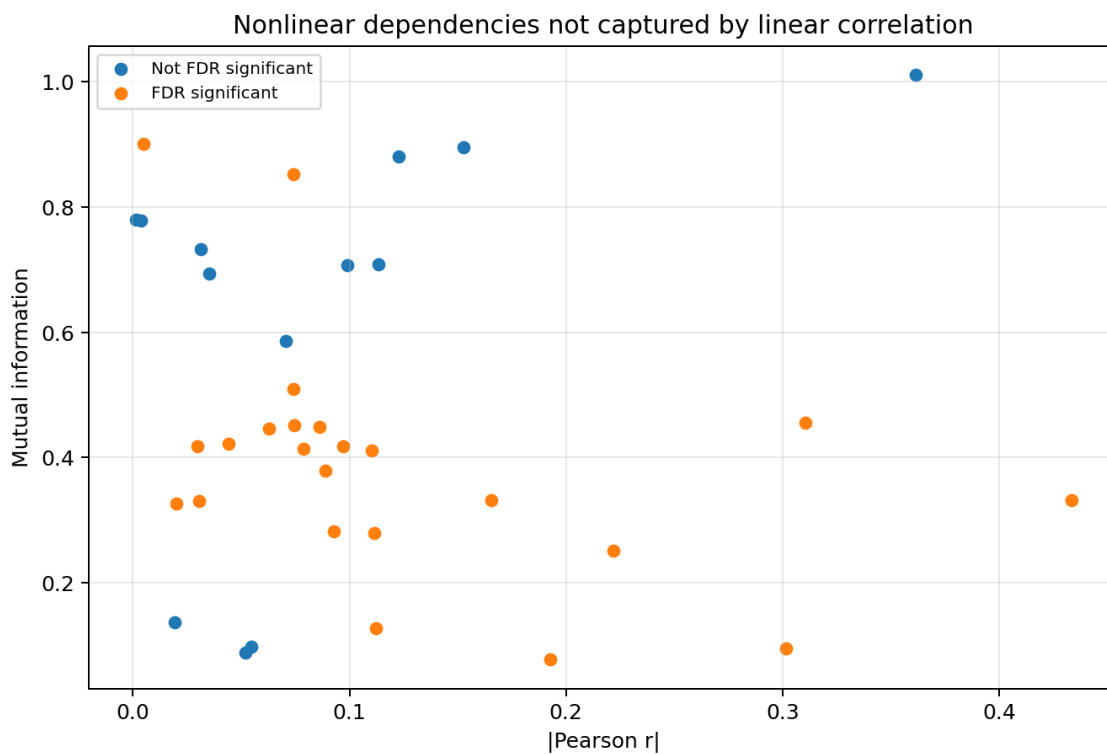


Figure 14. Mutual information versus absolute Pearson correlation for screened channel pairs. Many FDR-significant dependencies lie away from the high-linear-correlation region.

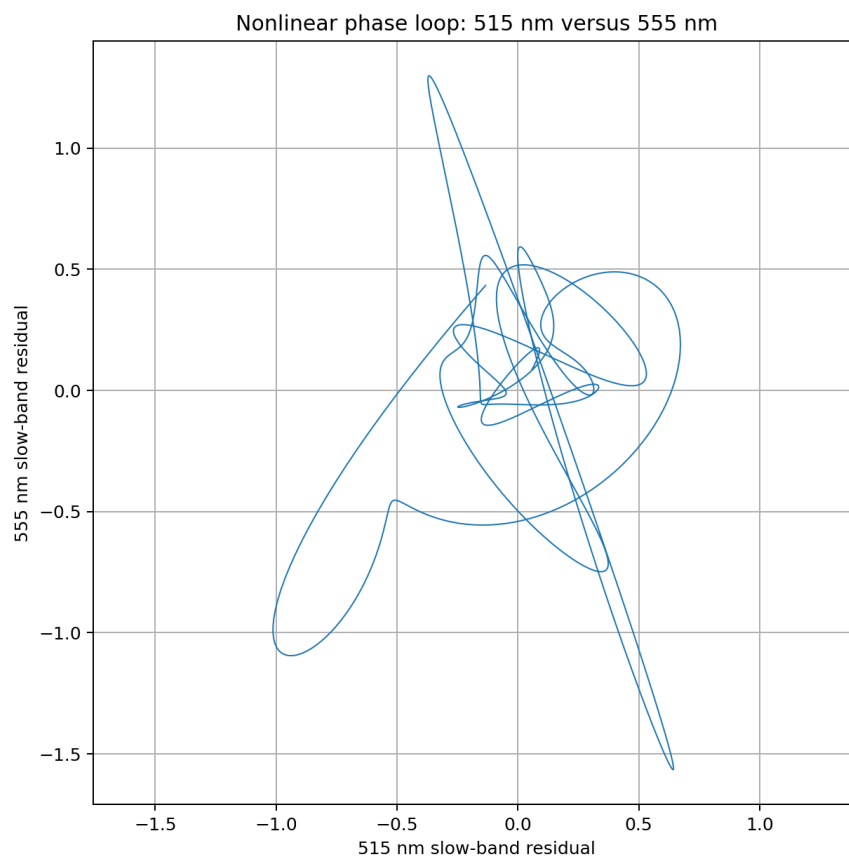


Figure 15. The 515-555 nm slow-band phase loop. Pearson r is approximately zero because the channels are near quadrature, but mutual information is high and survives FDR control.

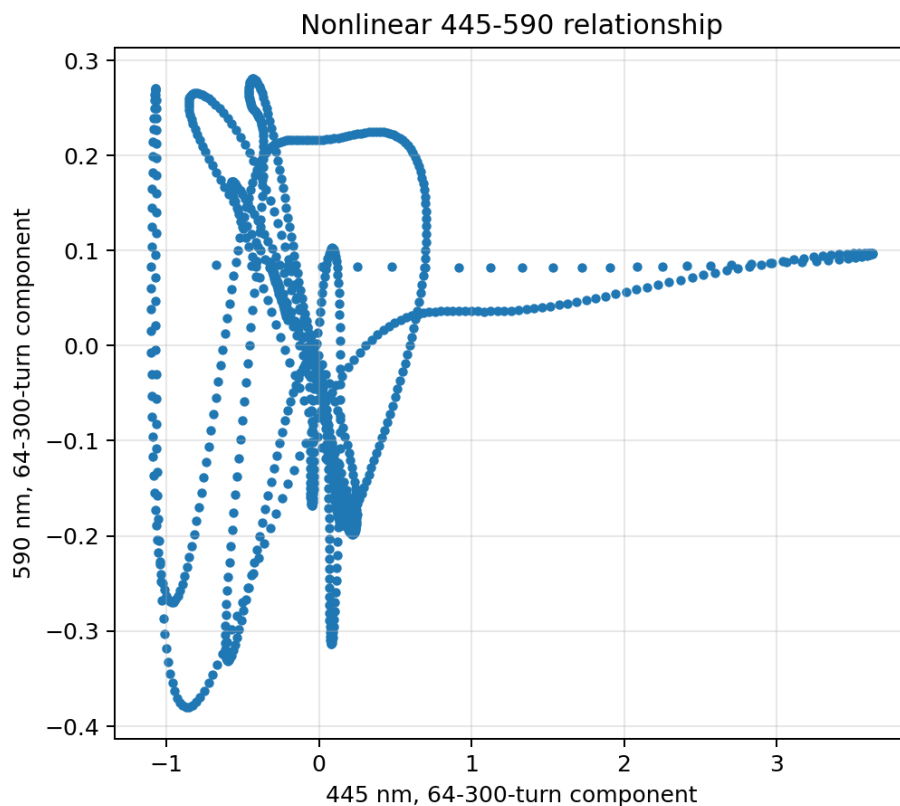


Figure 16. The 445-590 nm slow-band relation. Its dependence is strong by mutual information but is not represented by one fixed linear slope.

Pair	Band	Pearson	Mutual information	MI null percentile	Phase relation	FDR q
515-555	64-300 turns	-0.004975	0.899636	98.4375%	90.9977 deg	0.0364
445-590	64-300 turns	0.074 (reported screen)	0.852 (reported screen)	100.0%	time-varying	0.0133
555-clear	64-300 turns	0.003792	0.777903	92.9688%	89.1922 deg	not below 0.05

Across the complete relationship screen, 23 of 36 mutual-information tests survive 5% FDR control. Across all three period bands, 78 of 135 complex-coherence tests survive the same control. The channel system is therefore a statistically dense coupled state, not a collection of independent detector bands.

11. Key Pairwise Relationships by Timescale

Band	Pair	Pearson	Coherence	Phase deg	Coh. null pct	MI	MI null pct
short-3-12	480-630	+0.753304	0.755194	-4.054	100.000%	0.792963	
short-3-12	515-680	+0.766585	0.767084	+2.067	100.000%	0.908347	
short-3-12	445-clear	+0.445899	0.445920	+0.549	100.000%	0.122528	
short-3-12	415-555	+0.310522	0.318226	+12.633	100.000%	0.455152	100.000%
mid-12-64	515-680	+0.882145	0.882198	-0.633	100.000%	1.071985	
mid-12-64	480-NIR	-0.694381	0.694427	+179.341	100.000%	0.468395	
mid-12-64	445-480	+0.568917	0.569048	-1.229	100.000%	0.434251	
mid-12-64	415-NIR	-0.563910	0.571207	+170.832	100.000%	0.341166	
slow-64-300	445-480	+0.964747	0.965551	+2.339	100.000%	1.810428	
slow-64-300	515-680	+0.961994	0.967965	+6.367	100.000%	1.880096	
slow-64-300	480-NIR	-0.958217	0.958225	+179.767	100.000%	1.699482	
slow-64-300	515-555	-0.004975	0.285730	+90.998	91.406%	0.899636	98.438%
slow-64-300	555-clear	+0.003792	0.269010	+89.192	69.531%	0.777903	92.969%

Two relationships are especially diagnostic. First, 480 and NIR approach perfect opposition at long periods: coherence 0.958 and phase 179.77 deg in the 64-300-turn band. Second, 515 and 680 approach an in-phase state: coherence 0.968 and phase 6.37 deg in the same band. These are direct expressions of the two-pole polarization.

12. Search for Additional Rectilinear Axes



Figure 17. Frequency, phase, and held-out phase performance for PC2 through PC5. Only PC2 combines a high-quality chirp with phase-coherent out-of-sample spatial closure.

Mode	Ridge points	Frequency R^2	Phase R^2	Holdout phase R^2	Spacing km	RA deg	Dec deg
PC2	45	0.949759	0.502897	0.504955	1566.144	291.732	+35.972
PC3	31	0.962611	0.071052	-0.025697	179.522	20.388	+73.085
PC4	15	0.432923	0.085709	-0.842928	613.926	147.106	-68.012
PC5	26	0.526884	0.158331	0.084350	366.530	236.028	-66.443

PC3 has a high frequency-track R^2 but only 0.071 phase closure and negative held-out phase performance. PC4 and PC5 also fail spatial phase closure. A frequency ridge alone is therefore insufficient evidence for a plane family. The dataset resolves one plane-stack axis, not a complete orthogonal three-axis rectilinear lattice.

13. Secondary ICRS Polarization Tests

The first device harmonic was also transformed into an oriented ICRS vector for each stack. This asks whether the angular response itself maintains a fixed inertial direction. The strongest raw first-harmonic registries are 515, 590, and 680 nm. A separate 31-sample angular-standard-deviation observable gives the strongest result at 515 nm.

Observable	Channel	Coherence	Null percentile	RA deg	Dec deg	Null max
raw_m1	515	0.190430	95.312%	289.247	-54.535	0.200411
raw_m1	590	0.196796	95.312%	268.110	-52.654	0.198842
raw_m1	680	0.176220	97.656%	272.271	-31.015	0.185868
std31_m1	515	0.175980	99.219%	279.887	+16.695	0.176514

These directions change between the raw harmonic and variance observables, and no second-harmonic tensor exceeds the 95th percentile. They are therefore secondary evidence. The strongest and most internally validated geometric result remains the translational plane phase derived from the unwrapped trajectory.

14. Geometric and Physical Interpretation

14.1 Rectilinear imprint

A rectilinear imprint means that the measured channel state repeats when the detector trajectory advances by a fixed distance along one inertial normal. The present result satisfies four independent conditions: the period changes as the projected velocity changes; one fixed q explains that change; the integrated spatial phase predicts held-out signal; and the phase organizes channel reversals. These conditions are stronger than a visual periodicity in time.

14.2 Polarization

Polarization here denotes a signed spectral state, not an assumption about conventional optical polarization. When the trajectory advances through half a plane cycle, 415-480 nm reverse relative to 515-680 nm and NIR. The phase difference of 175.76 deg is close to ideal opposition. In the multichannel state space, this becomes the second principal component after the broadband common mode.

14.3 Relationship to spherical coherence

The earlier unit-sphere analysis established broad degree-2 directional structure. The unwrapped analysis shows why a sphere alone is incomplete: Earth translation carries the detector through new spatial phase even when local orientation repeats. A spherical map can represent orientation dependence, while the unwrapped trajectory captures translation through planes. The two descriptions are complementary. The resolved plane wave acts on position; the device-angle harmonics and spherical modes describe how the rotating probe samples that field.

14.4 Limits of the present result

The data establish one primitive spacing and one inertial normal. Tangential periods remain unconstrained. The five-turn profiles overlap strongly, so neighboring stacks are not independent. Circular-shift nulls and held-out tests account for this autocorrelation, but an independent replication with a different start epoch, rotation direction, or laboratory trajectory would provide a stronger geometric test. The 147 flagged rows were retained; a future instrument-health analysis should predefine exclusion rules. None of these limitations changes the measured one-axis phase closure, but they define the boundary of the present claim.

15. Reproduction Procedure

1. Read `pic_id_14C.csv` and compute effective UTC as `date.UTC` minus `frame_read_us`.
2. Partition the record into complete 90-index revolutions with indices 0 through 89.
3. Form each five-turn profile by elementwise summation of five consecutive revolutions; advance by one revolution.
4. Register every profile to the time and ICRS basis of its center revolution.
5. Compute $R(t, \theta)$ from CMB translation, Earth-rotating lab position, and detector offset.
6. For each channel, calculate the logarithmic five-turn DC series and the 3-12, 12-64, and 64-300-turn components.
7. Standardize the channel matrix within each band and calculate PCA.
8. Extract the 12-64-turn PC2 frequency ridge using the declared sliding window and period search.
9. Fit q from observed ridge frequency to $q \cdot v(t)$; calculate $d = 1/|q|$ and the normal direction.
10. Integrate spatial phase and fit sinusoidal phase models to PC2, channel residuals, and blue-minus-red/NIR contrast.
11. Validate with held-out phase prediction, even/odd ridge splits, circular shifts, reversal matching, and sensitivity grids.
12. Compute device-angle complex harmonics, channel-pair matching, mutual information, distance correlation, and FDR corrections.

16. Conclusions

- The laboratory traversed approximately 4.01×10^{10} m along the supplied CMB vector during a record spanning 180.02 deg of Earth rotation.
- The multichannel response is low dimensional and separates into a broadband common mode and a blue-versus-red/NIR polarized mode.
- The polarized mode contains a velocity-governed chirp consistent with a fixed inertial stack of parallel planes.
- The resolved primitive spacing is 1,566.144 km; sign-reversal sheets occur every 783.072 km.
- The plane normal is RA 291.732 deg, Dec +35.972 deg, nearly perpendicular to the CMB direction.

- The fixed geometry explains 94.98% of the frequency ridge and approximately 60% of held-out spatial phase.
- The blue and red/NIR phase groups are separated by 175.76 deg, and 24 of 29 fitted reversals are matched within five revolutions.
- Device-angle harmonics reveal the stable pairs 415-NIR, 445-clear, 480-630, 515-680, and 555-590.
- Nonlinear dependencies are widespread; 23 of 36 screened mutual-information tests and 78 of 135 phase-coherence tests survive 5% FDR control.
- No second independent plane family achieves spatial phase closure. A full three-dimensional rectilinear cell is therefore not yet established.

Appendix A. Complete Pairwise Band-Relationship Ledger

The following table contains all 135 channel-pair tests across the three period bands. Mutual-information fields are blank where the pair was not selected for the nonlinear screen.

Band	Pair	Pearson	Spearman	dCor	Coherence	Phase deg	Null pct	MI	MI FDR q
short-3-12	415-445	-0.2551	-0.2647	0.2605	0.2584	-170.81	96.9%	0.0819	
short-3-12	415-480	-0.1842	-0.0777	0.1634	0.2260	+144.57	100.0%	0.4498	
short-3-12	415-515	-0.0442	+0.0339	0.1326	0.0990	-116.54	51.6%	0.4221	0.0133
short-3-12	415-555	+0.3105	+0.3772	0.3683	0.3182	+12.63	100.0%	0.4552	0.0133
short-3-12	415-590	-0.0888	-0.0127	0.1163	0.0923	-164.03	49.2%	0.3789	0.0133
short-3-12	415-630	-0.1201	-0.0824	0.1335	0.1758	+133.09	96.1%	0.4111	
short-3-12	415-680	-0.1101	-0.0297	0.1361	0.1279	-149.41	81.2%	0.4118	0.0133
short-3-12	415-NIR	-0.1926	-0.2374	0.2341	0.1928	-177.17	96.9%	0.0786	0.0133
short-3-12	415-clear	-0.3040	-0.3242	0.3104	0.3175	-163.27	100.0%	0.0955	
short-3-12	445-480	-0.3812	-0.3956	0.3728	0.3925	+166.19	100.0%	0.1421	
short-3-12	445-515	-0.3553	-0.3601	0.3430	0.3739	-161.88	100.0%	0.1507	
short-3-12	445-555	-0.2846	-0.2592	0.2638	0.2885	+170.62	100.0%	0.0883	
short-3-12	445-590	+0.2220	+0.2996	0.3357	0.2246	+8.70	99.2%	0.2510	0.0133
short-3-12	445-630	-0.3719	-0.3711	0.3566	0.3813	+167.22	100.0%	0.1307	
short-3-12	445-680	-0.3668	-0.3662	0.3495	0.4093	-153.66	100.0%	0.1469	
short-3-12	445-NIR	+0.2675	+0.1460	0.1902	0.2676	-1.84	100.0%	0.0468	
short-3-12	445-clear	+0.4459	+0.2903	0.3146	0.4459	+0.55	100.0%	0.1225	
short-3-12	480-515	-0.2072	-0.1269	0.1896	0.2343	+152.16	97.7%	0.5009	
short-3-12	480-555	-0.0746	+0.0093	0.1364	0.0860	-150.19	60.2%	0.4515	0.0133
short-3-12	480-590	-0.1604	-0.0661	0.1594	0.1788	-153.81	91.4%	0.4503	
short-3-12	480-630	+0.7533	+0.7345	0.7091	0.7552	-4.05	100.0%	0.7930	
short-3-12	480-680	-0.1751	-0.0981	0.1593	0.2550	+133.38	99.2%	0.4580	
short-3-12	480-NIR	-0.3014	-0.3357	0.3193	0.3049	+171.31	100.0%	0.0960	0.0133
short-3-12	480-clear	-0.2552	-0.2662	0.2694	0.2600	+169.00	100.0%	0.1177	
short-3-12	515-555	-0.0297	+0.0114	0.0995	0.0330	+154.40	8.6%	0.4186	0.0133
short-3-12	515-590	-0.1782	-0.1167	0.1642	0.1853	+164.14	99.2%	0.4817	
short-3-12	515-630	-0.3126	-0.2168	0.2702	0.3179	-169.55	100.0%	0.5426	
short-3-12	515-680	+0.7666	+0.7810	0.7504	0.7671	+2.07	100.0%	0.9083	
short-3-12	515-NIR	-0.2607	-0.2637	0.2603	0.2742	-161.99	100.0%	0.0935	
short-3-12	515-clear	-0.2818	-0.2771	0.2739	0.2828	-175.21	100.0%	0.1067	
short-3-12	555-590	+0.0629	+0.1309	0.1755	0.0677	-21.77	21.9%	0.4458	0.0133
short-3-12	555-630	-0.0970	-0.0088	0.1302	0.1183	+145.08	83.6%	0.4177	0.0133
short-3-12	555-680	-0.1193	-0.0625	0.1371	0.1219	-168.00	67.2%	0.4595	
short-3-12	555-NIR	-0.3374	-0.3191	0.3059	0.3377	+177.91	100.0%	0.0909	
short-3-12	555-clear	-0.2253	-0.2256	0.2287	0.2312	+167.02	100.0%	0.0896	
short-3-12	590-630	-0.0862	+0.0070	0.1300	0.1295	+131.74	83.6%	0.4489	0.0133
short-3-12	590-680	-0.1806	-0.1310	0.1611	0.2440	-137.72	100.0%	0.4949	
short-3-12	590-NIR	-0.2023	-0.2072	0.2166	0.2078	+166.73	98.4%	0.0762	
short-3-12	590-clear	-0.1983	-0.1897	0.2016	0.2018	-169.41	97.7%	0.0770	
short-3-12	630-680	-0.2189	-0.1287	0.1928	0.2551	+149.11	100.0%	0.5231	
short-3-12	630-NIR	-0.2391	-0.2648	0.2540	0.2712	+151.86	100.0%	0.0648	
short-3-12	630-clear	-0.2996	-0.3120	0.3072	0.3003	-175.96	100.0%	0.1351	
short-3-12	680-NIR	-0.2284	-0.2295	0.2258	0.2396	-162.45	100.0%	0.0915	
short-3-12	680-clear	-0.2847	-0.2943	0.2872	0.2851	+176.86	100.0%	0.1162	
short-3-12	NIR-clear	+0.2415	+0.1438	0.1648	0.2471	+12.21	99.2%	0.0531	
mid-12-64	415-445	+0.4207	+0.3865	0.4171	0.4313	-12.71	100.0%	0.2596	
mid-12-64	415-480	+0.6155	+0.5590	0.6176	0.6166	-3.50	100.0%	0.5847	
mid-12-64	415-515	-0.3170	-0.2889	0.3074	0.3202	-171.85	100.0%	0.3199	
mid-12-64	415-555	-0.2213	-0.0201	0.2003	0.3226	+133.33	92.2%	0.3164	
mid-12-64	415-590	-0.3686	-0.3175	0.3463	0.3686	+179.10	99.2%	0.3224	
mid-12-64	415-630	-0.3467	-0.3108	0.3284	0.3522	+169.88	100.0%	0.2785	
mid-12-64	415-680	-0.3544	-0.3066	0.3296	0.3549	-177.12	100.0%	0.3055	
mid-12-64	415-NIR	-0.5639	-0.5964	0.5826	0.5712	+170.83	100.0%	0.3412	
mid-12-64	415-clear	-0.1580	-0.0961	0.1557	0.2483	-129.51	93.0%	0.1410	
mid-12-64	445-480	+0.5689	+0.4499	0.5603	0.5690	-1.23	100.0%	0.4343	
mid-12-64	445-515	-0.5469	-0.5203	0.5086	0.5525	-171.81	100.0%	0.2904	
mid-12-64	445-555	-0.5832	-0.5770	0.5488	0.5941	+169.00	100.0%	0.3150	
mid-12-64	445-590	-0.1876	-0.1856	0.2256	0.2069	+155.06	83.6%	0.2079	

Band	Pair	Pearson	Spearman	dCor	Coherence	Phase deg	Null pct	MI	MI FDR q
mid-12-64	445-630	-0.4684	-0.4508	0.4278	0.4697	-175.78	100.0%	0.1852	
mid-12-64	445-680	-0.5816	-0.5641	0.5418	0.5821	-177.56	100.0%	0.2972	
mid-12-64	445-NIR	-0.4333	-0.4899	0.4971	0.4341	+176.33	100.0%	0.3315	0.0133
mid-12-64	445-clear	+0.3845	+0.3150	0.3224	0.4233	-24.72	100.0%	0.1359	
mid-12-64	480-515	-0.4426	-0.3105	0.4001	0.4527	-167.87	100.0%	0.4012	
mid-12-64	480-555	-0.4283	-0.3510	0.4063	0.4398	+166.84	100.0%	0.4210	
mid-12-64	480-590	-0.3101	-0.1718	0.2854	0.3840	+143.85	100.0%	0.3876	
mid-12-64	480-630	-0.0740	+0.0205	0.1803	0.0744	-174.48	21.1%	0.5099	0.0133
mid-12-64	480-680	-0.5044	-0.3451	0.4378	0.5046	+178.67	100.0%	0.3682	
mid-12-64	480-NIR	-0.6944	-0.6858	0.6611	0.6944	+179.34	100.0%	0.4684	
mid-12-64	480-clear	+0.0194	-0.0093	0.1254	0.2418	-85.41	86.7%	0.1372	0.1296
mid-12-64	515-555	+0.0929	+0.0830	0.1551	0.1596	-54.40	65.6%	0.2830	0.0133
mid-12-64	515-590	+0.1656	+0.1742	0.2141	0.1868	-27.56	84.4%	0.3318	0.0133
mid-12-64	515-630	-0.0199	+0.0106	0.1404	0.0394	-120.36	10.9%	0.3274	0.0133
mid-12-64	515-680	+0.8821	+0.8727	0.8505	0.8822	-0.63	100.0%	1.0720	
mid-12-64	515-NIR	+0.1124	+0.1316	0.1694	0.1147	-11.39	57.0%	0.1274	0.0254
mid-12-64	515-clear	-0.3674	-0.3207	0.3439	0.3687	+175.32	100.0%	0.2094	
mid-12-64	555-590	+0.1114	+0.1153	0.1654	0.1606	-46.07	60.2%	0.2804	0.0133
mid-12-64	555-630	+0.1537	+0.1618	0.1910	0.1795	+31.09	77.3%	0.3141	
mid-12-64	555-680	+0.1954	+0.1936	0.1980	0.2165	+25.51	81.2%	0.2260	
mid-12-64	555-NIR	+0.1472	+0.1516	0.1675	0.1839	+36.84	66.4%	0.0957	
mid-12-64	555-clear	-0.2786	-0.2987	0.2999	0.3260	+148.74	96.1%	0.1542	
mid-12-64	590-630	+0.0788	+0.1927	0.2435	0.0988	-37.08	31.2%	0.4142	0.0133
mid-12-64	590-680	+0.2063	+0.2197	0.2407	0.2255	+23.83	90.6%	0.3290	
mid-12-64	590-NIR	+0.0520	+0.0687	0.1104	0.0620	+32.96	23.4%	0.0887	0.5145
mid-12-64	590-clear	-0.3408	-0.3149	0.3160	0.3475	+168.74	100.0%	0.1474	
mid-12-64	630-680	-0.0305	+0.0091	0.1325	0.0339	+154.05	3.1%	0.3311	0.0133
mid-12-64	630-NIR	+0.1708	+0.1475	0.1828	0.1751	-12.58	89.1%	0.1215	
mid-12-64	630-clear	-0.3300	-0.3661	0.3430	0.3377	+167.76	100.0%	0.1545	
mid-12-64	680-NIR	+0.1303	+0.1225	0.1740	0.1442	+25.33	58.6%	0.1072	
mid-12-64	680-clear	-0.3712	-0.3185	0.3366	0.3747	+172.14	100.0%	0.2168	
mid-12-64	NIR-clear	+0.0547	-0.0861	0.1714	0.2531	+77.52	90.6%	0.0985	0.2698
slow-64-300	415-445	+0.8935	+0.7506	0.8326	0.9076	-10.11	100.0%	1.4072	
slow-64-300	415-480	+0.8590	+0.6136	0.8019	0.8613	-4.19	100.0%	1.5385	
slow-64-300	415-515	-0.4329	-0.1580	0.4681	0.5675	+139.72	100.0%	1.0407	
slow-64-300	415-555	-0.6597	-0.5428	0.6192	0.6813	-165.55	100.0%	1.1165	
slow-64-300	415-590	+0.0351	-0.0192	0.3181	0.1055	-70.60	0.8%	0.6928	0.6895
slow-64-300	415-630	-0.3344	-0.3575	0.3968	0.4037	+145.93	75.0%	0.9272	
slow-64-300	415-680	-0.5912	-0.2307	0.5745	0.6472	+155.99	100.0%	1.2394	
slow-64-300	415-NIR	-0.7397	-0.4165	0.6717	0.7490	+170.97	100.0%	1.3972	
slow-64-300	415-clear	+0.2753	+0.1624	0.3171	0.5050	-56.97	90.6%	0.7975	
slow-64-300	445-480	+0.9647	+0.7619	0.9204	0.9656	+2.34	100.0%	1.8104	
slow-64-300	445-515	-0.7170	-0.5356	0.6524	0.7759	+157.54	100.0%	1.0096	
slow-64-300	445-555	-0.5381	-0.5204	0.5581	0.5810	-157.83	100.0%	0.8985	
slow-64-300	445-590	+0.0740	-0.1063	0.3258	0.2370	-71.81	28.9%	0.8521	0.0133
slow-64-300	445-630	-0.2889	-0.3890	0.3574	0.2948	+168.49	57.8%	0.8979	
slow-64-300	445-680	-0.8220	-0.5762	0.7466	0.8382	+168.72	100.0%	1.1323	
slow-64-300	445-NIR	-0.9218	-0.7222	0.8444	0.9218	+179.98	100.0%	1.3096	
slow-64-300	445-clear	+0.5646	+0.6087	0.5662	0.6215	-24.70	100.0%	0.9089	
slow-64-300	480-515	-0.7087	-0.5162	0.6508	0.7577	+159.28	100.0%	1.0318	
slow-64-300	480-555	-0.5195	-0.4776	0.5874	0.5266	-170.53	100.0%	1.1509	
slow-64-300	480-590	+0.1133	+0.0681	0.2636	0.2911	-67.09	44.5%	0.7083	0.7654
slow-64-300	480-630	-0.1565	-0.0991	0.2175	0.1571	+174.85	28.1%	0.8913	
slow-64-300	480-680	-0.8276	-0.5730	0.7593	0.8417	+169.52	100.0%	1.1515	
slow-64-300	480-NIR	-0.9582	-0.8626	0.9106	0.9582	+179.77	100.0%	1.6995	
slow-64-300	480-clear	+0.4646	+0.4055	0.4448	0.4929	-19.51	100.0%	0.9000	
slow-64-300	515-555	-0.0050	-0.0380	0.3475	0.2857	+91.00	91.4%	0.8996	0.0364
slow-64-300	515-590	-0.3523	-0.3529	0.3581	0.5117	+133.51	89.8%	0.6873	
slow-64-300	515-630	+0.0312	+0.0357	0.2023	0.1478	+77.81	0.0%	0.7321	0.5666
slow-64-300	515-680	+0.9620	+0.9615	0.9520	0.9680	+6.37	100.0%	1.8801	
slow-64-300	515-NIR	+0.7519	+0.6603	0.7111	0.7921	+18.33	100.0%	1.0748	
slow-64-300	515-clear	-0.6144	-0.5420	0.5531	0.6314	-166.67	100.0%	0.8495	
slow-64-300	555-590	+0.0989	+0.1908	0.2549	0.0996	-6.94	25.8%	0.7069	0.1828

Band	Pair	Pearson	Spearman	dCor	Coherence	Phase deg	Null pct	MI	MI FDR q
slow-64-300	555-630	-0.1226	-0.1097	0.2502	0.2132	-125.11	9.4%	0.8800	0.0558
slow-64-300	555-680	+0.1526	+0.0437	0.3795	0.2659	-54.99	80.5%	0.8947	0.1034
slow-64-300	555-NIR	+0.3614	+0.2436	0.4896	0.3784	-17.25	93.0%	1.0102	0.0558
slow-64-300	555-clear	+0.0038	-0.0951	0.2790	0.2690	+89.19	69.5%	0.7779	0.1034
slow-64-300	590-630	+0.1888	+0.1553	0.2268	0.2095	+25.69	38.3%	0.6622	
slow-64-300	590-680	-0.2962	-0.3055	0.3339	0.4568	-130.42	82.0%	0.7392	
slow-64-300	590-NIR	-0.0705	+0.0276	0.2371	0.3404	-101.95	57.0%	0.5854	0.9070
slow-64-300	590-clear	-0.2856	-0.2642	0.3399	0.4023	+135.22	79.7%	0.7539	
slow-64-300	630-680	+0.0014	-0.0226	0.2070	0.1177	-89.33	0.8%	0.7789	0.5131
slow-64-300	630-NIR	+0.1645	+0.1408	0.2638	0.1703	-15.03	14.1%	0.9381	
slow-64-300	630-clear	-0.5222	-0.5551	0.5333	0.6312	+145.83	100.0%	0.8576	
slow-64-300	680-NIR	+0.8394	+0.6917	0.7772	0.8504	+9.20	100.0%	1.0841	
slow-64-300	680-clear	-0.5445	-0.4806	0.5067	0.5472	-174.30	100.0%	0.8839	
slow-64-300	NIR-clear	-0.5636	-0.5651	0.5349	0.5736	+169.27	100.0%	0.8443	

Appendix B. Complete Device-Angle Harmonic Ledger

All 180 channel-pair harmonic tests are listed below. Null percentiles use autocorrelation-preserving circular shifts.

Harmonic	Pair	Coherence	Phase deg	Null percentile	Null mean	Null max
1	415-445	0.463749	+3.657	100.000%	0.051564	0.118846
1	415-480	0.490080	+4.673	100.000%	0.045060	0.135647
1	415-515	0.447202	+5.410	100.000%	0.053404	0.138802
1	415-555	0.565942	+3.052	100.000%	0.052020	0.122134
1	415-590	0.536847	+3.900	100.000%	0.051063	0.136498
1	415-630	0.483579	-2.527	100.000%	0.054044	0.128415
1	415-680	0.502923	-0.876	100.000%	0.046845	0.160879
1	415-NIR	0.372950	+0.544	100.000%	0.054252	0.128585
1	415-clear	0.453494	+0.852	100.000%	0.050665	0.139947
1	445-480	0.573174	-1.758	100.000%	0.049331	0.177932
1	445-515	0.538209	+0.778	100.000%	0.053503	0.155496
1	445-555	0.615662	+1.891	100.000%	0.054073	0.148403
1	445-590	0.688894	+0.403	100.000%	0.051393	0.145863
1	445-630	0.582144	-1.369	100.000%	0.052966	0.149682
1	445-680	0.543296	-2.844	100.000%	0.055388	0.123313
1	445-NIR	0.520632	-2.915	100.000%	0.057359	0.123080
1	445-clear	0.701264	+0.884	100.000%	0.058355	0.167010
1	480-515	0.557931	+4.535	100.000%	0.053599	0.121138
1	480-555	0.639772	+3.319	100.000%	0.051896	0.163405
1	480-590	0.672108	+1.380	100.000%	0.054854	0.132939
1	480-630	0.724214	+2.291	100.000%	0.048661	0.114437
1	480-680	0.565032	-2.970	100.000%	0.058956	0.142907
1	480-NIR	0.409336	+1.386	100.000%	0.052865	0.128316
1	480-clear	0.546973	+2.463	100.000%	0.053009	0.132772
1	515-555	0.585415	-2.133	100.000%	0.061594	0.145163
1	515-590	0.653738	-2.278	100.000%	0.060491	0.156934
1	515-630	0.591702	-1.984	100.000%	0.057173	0.145120
1	515-680	0.711428	+0.700	100.000%	0.065854	0.127962
1	515-NIR	0.451146	-1.988	100.000%	0.060368	0.169748
1	515-clear	0.534681	-1.477	100.000%	0.063686	0.138338
1	555-590	0.763762	-1.131	100.000%	0.054844	0.151807
1	555-630	0.678818	-3.222	100.000%	0.055800	0.153039
1	555-680	0.655759	-4.963	100.000%	0.057352	0.146422
1	555-NIR	0.461054	-5.268	100.000%	0.059923	0.126701
1	555-clear	0.608643	-1.120	100.000%	0.060935	0.135875
1	590-630	0.714366	-3.326	100.000%	0.054210	0.139563
1	590-680	0.682536	-2.004	100.000%	0.062388	0.146573
1	590-NIR	0.503765	-3.848	100.000%	0.059886	0.129900
1	590-clear	0.650434	+1.639	100.000%	0.059922	0.143218
1	630-680	0.593852	+0.805	100.000%	0.057671	0.132506
1	630-NIR	0.471261	+1.780	100.000%	0.057764	0.138210
1	630-clear	0.576838	+1.561	100.000%	0.056532	0.172877
1	680-NIR	0.465957	-2.251	100.000%	0.061656	0.140252
1	680-clear	0.586398	+2.895	100.000%	0.061991	0.140132
1	NIR-clear	0.524908	-0.095	100.000%	0.070526	0.155365
2	415-445	0.339463	-3.926	100.000%	0.052592	0.114836
2	415-480	0.306565	-11.231	100.000%	0.047560	0.116194
2	415-515	0.371766	-6.534	100.000%	0.050314	0.154775
2	415-555	0.423846	-9.130	100.000%	0.045772	0.134753
2	415-590	0.366200	-4.333	100.000%	0.050023	0.125113
2	415-630	0.342967	-11.708	100.000%	0.047970	0.147291
2	415-680	0.320027	-15.271	100.000%	0.048500	0.143502
2	415-NIR	0.193852	-2.952	100.000%	0.052023	0.114936
2	415-clear	0.304989	-8.093	100.000%	0.055762	0.127124
2	445-480	0.489474	-5.788	100.000%	0.052799	0.133886
2	445-515	0.463969	-4.932	100.000%	0.056059	0.137710
2	445-555	0.528179	-2.340	100.000%	0.056577	0.164010
2	445-590	0.586290	-1.197	100.000%	0.050759	0.129766
2	445-630	0.477258	-5.136	100.000%	0.054596	0.117308

Harmonic	Pair	Coherence	Phase deg	Null percentile	Null mean	Null max
2	445-680	0.440670	-6.100	100.000%	0.055520	0.183388
2	445-NIR	0.373391	-11.983	100.000%	0.054932	0.125484
2	445-clear	0.640535	-7.290	100.000%	0.057347	0.141657
2	480-515	0.391460	+2.105	100.000%	0.052501	0.130173
2	480-555	0.470549	-2.463	100.000%	0.049213	0.128204
2	480-590	0.474228	-0.770	100.000%	0.050600	0.107577
2	480-630	0.520604	-3.968	100.000%	0.048795	0.133988
2	480-680	0.426123	-1.000	100.000%	0.050837	0.116516
2	480-NIR	0.273985	-1.718	100.000%	0.053079	0.130777
2	480-clear	0.436454	+4.080	100.000%	0.060984	0.136796
2	515-555	0.506258	-0.537	100.000%	0.048155	0.138868
2	515-590	0.496326	-3.821	100.000%	0.049436	0.128584
2	515-630	0.459496	-1.703	100.000%	0.051161	0.130303
2	515-680	0.581389	-3.669	100.000%	0.053207	0.157248
2	515-NIR	0.274324	-8.615	100.000%	0.055360	0.138137
2	515-clear	0.444829	-0.270	100.000%	0.057050	0.130643
2	555-590	0.569354	+4.197	100.000%	0.045084	0.112962
2	555-630	0.521009	-2.315	100.000%	0.047057	0.114658
2	555-680	0.499263	-2.254	100.000%	0.049188	0.124349
2	555-NIR	0.303245	-3.875	100.000%	0.050899	0.140324
2	555-clear	0.505730	+3.591	100.000%	0.054500	0.169183
2	590-630	0.548816	-3.755	100.000%	0.048083	0.117850
2	590-680	0.489818	-3.310	100.000%	0.050008	0.139027
2	590-NIR	0.357145	-9.739	100.000%	0.049902	0.140697
2	590-clear	0.536270	-3.333	100.000%	0.053950	0.159199
2	630-680	0.450724	+1.030	100.000%	0.050544	0.128502
2	630-NIR	0.377771	-2.919	100.000%	0.055526	0.169311
2	630-clear	0.495724	+2.186	100.000%	0.055383	0.141970
2	680-NIR	0.333035	+2.188	100.000%	0.054991	0.164937
2	680-clear	0.474880	-4.005	100.000%	0.053002	0.131100
2	NIR-clear	0.400480	+6.762	100.000%	0.051667	0.156511
3	415-445	0.240189	+6.093	100.000%	0.052474	0.139937
3	415-480	0.334647	+11.060	100.000%	0.050772	0.128900
3	415-515	0.268069	-9.041	100.000%	0.049787	0.188134
3	415-555	0.238886	-21.457	100.000%	0.053403	0.134238
3	415-590	0.304179	+6.283	100.000%	0.057404	0.136703
3	415-630	0.305964	+0.963	100.000%	0.051377	0.155527
3	415-680	0.291730	-3.065	100.000%	0.051591	0.124008
3	415-NIR	0.143373	+5.066	96.875%	0.058181	0.176460
3	415-clear	0.261713	-8.670	100.000%	0.053412	0.123401
3	445-480	0.382003	+2.743	100.000%	0.052032	0.135921
3	445-515	0.380335	+0.207	100.000%	0.054904	0.129774
3	445-555	0.407298	+2.680	100.000%	0.053426	0.123868
3	445-590	0.487433	-1.121	100.000%	0.056341	0.124046
3	445-630	0.373525	+0.205	100.000%	0.054124	0.129980
3	445-680	0.393807	-1.758	100.000%	0.055343	0.157894
3	445-NIR	0.253350	-9.707	100.000%	0.052271	0.149072
3	445-clear	0.511577	-5.981	100.000%	0.053950	0.126896
3	480-515	0.407447	-9.671	100.000%	0.052888	0.133916
3	480-555	0.411877	-11.230	100.000%	0.051335	0.163514
3	480-590	0.407278	-4.996	100.000%	0.051305	0.140924
3	480-630	0.437841	-2.911	100.000%	0.057634	0.119569
3	480-680	0.391045	-5.999	100.000%	0.051581	0.114240
3	480-NIR	0.170438	-16.570	100.000%	0.059878	0.137664
3	480-clear	0.396282	-11.943	100.000%	0.052655	0.126837
3	515-555	0.343372	+2.249	100.000%	0.055044	0.118486
3	515-590	0.419495	-6.209	100.000%	0.052860	0.142048
3	515-630	0.306648	-0.925	100.000%	0.050956	0.142015
3	515-680	0.468386	-7.251	100.000%	0.053514	0.144031
3	515-NIR	0.123916	-22.197	98.438%	0.050801	0.130425
3	515-clear	0.376426	-1.415	100.000%	0.052193	0.162109
3	555-590	0.480899	-2.652	100.000%	0.052876	0.125928
3	555-630	0.405066	-0.105	100.000%	0.051588	0.146066

Harmonic	Pair	Coherence	Phase deg	Null percentile	Null mean	Null max
3	555-680	0.429040	-3.757	100.000%	0.057274	0.145364
3	555-NIR	0.197496	-10.375	100.000%	0.053554	0.124305
3	555-clear	0.430497	-6.654	100.000%	0.057349	0.143209
3	590-630	0.418913	-3.706	100.000%	0.051772	0.148535
3	590-680	0.468723	+3.887	100.000%	0.054955	0.136306
3	590-NIR	0.210180	-26.825	100.000%	0.051303	0.123983
3	590-clear	0.458656	-1.589	100.000%	0.053937	0.135255
3	630-680	0.374180	+6.439	100.000%	0.052332	0.130911
3	630-NIR	0.232955	-10.428	100.000%	0.053682	0.121125
3	630-clear	0.377866	-5.537	100.000%	0.053261	0.141359
3	680-NIR	0.224346	-7.200	100.000%	0.050249	0.150417
3	680-clear	0.336517	-7.291	100.000%	0.051151	0.134030
3	NIR-clear	0.309410	-3.681	100.000%	0.056029	0.126632
4	415-445	0.254153	+35.670	100.000%	0.056301	0.141111
4	415-480	0.212796	+14.605	100.000%	0.059558	0.158190
4	415-515	0.149170	+0.858	100.000%	0.052230	0.127718
4	415-555	0.238062	-4.111	100.000%	0.055884	0.144680
4	415-590	0.225395	+7.314	100.000%	0.049334	0.130749
4	415-630	0.246884	+15.822	100.000%	0.052927	0.136244
4	415-680	0.179221	-0.912	100.000%	0.051067	0.117780
4	415-NIR	0.081424	+20.509	78.125%	0.056710	0.141095
4	415-clear	0.203593	+4.652	100.000%	0.055853	0.144242
4	445-480	0.334597	+0.085	100.000%	0.056937	0.129748
4	445-515	0.310024	+0.941	100.000%	0.057688	0.125559
4	445-555	0.319716	-11.949	100.000%	0.050033	0.124919
4	445-590	0.356181	+1.326	100.000%	0.055033	0.118341
4	445-630	0.292922	-16.203	100.000%	0.053921	0.130106
4	445-680	0.290086	-4.814	100.000%	0.047835	0.123848
4	445-NIR	0.241439	-11.013	100.000%	0.056277	0.145915
4	445-clear	0.414939	-12.440	100.000%	0.053743	0.141995
4	480-515	0.303522	-7.078	100.000%	0.051661	0.137712
4	480-555	0.426669	-4.254	100.000%	0.051740	0.137768
4	480-590	0.386499	-2.531	100.000%	0.049557	0.142222
4	480-630	0.315994	+4.347	100.000%	0.052362	0.122245
4	480-680	0.307477	-6.436	100.000%	0.049034	0.127513
4	480-NIR	0.111214	+31.802	95.312%	0.055395	0.141179
4	480-clear	0.320141	-11.195	100.000%	0.052065	0.139008
4	515-555	0.351928	+4.073	100.000%	0.057511	0.152298
4	515-590	0.298691	+6.040	100.000%	0.050762	0.123431
4	515-630	0.298110	+16.099	100.000%	0.049902	0.116917
4	515-680	0.381396	-4.127	100.000%	0.047435	0.143661
4	515-NIR	0.149123	+13.990	100.000%	0.057741	0.148196
4	515-clear	0.295143	+0.363	100.000%	0.057307	0.149908
4	555-590	0.480361	+5.323	100.000%	0.050996	0.113592
4	555-630	0.381295	+9.015	100.000%	0.058236	0.140526
4	555-680	0.394678	-10.517	100.000%	0.053748	0.177100
4	555-NIR	0.230788	+25.968	100.000%	0.060753	0.124424
4	555-clear	0.365068	+1.808	100.000%	0.058077	0.160016
4	590-630	0.398829	+0.279	100.000%	0.053703	0.143212
4	590-680	0.365166	-0.188	100.000%	0.046414	0.124472
4	590-NIR	0.207109	-2.990	100.000%	0.052475	0.168986
4	590-clear	0.369276	-5.152	100.000%	0.050549	0.139156
4	630-680	0.337697	-15.560	100.000%	0.048706	0.145713
4	630-NIR	0.217865	-1.954	100.000%	0.056964	0.151178
4	630-clear	0.333069	-6.094	100.000%	0.051433	0.117473
4	680-NIR	0.174029	-2.796	100.000%	0.052102	0.125632
4	680-clear	0.265581	+5.914	100.000%	0.051852	0.122447
4	NIR-clear	0.287702	+3.989	100.000%	0.052636	0.169492

Appendix C. Complete ICRS Polarization-Metric Ledger

Observable	Channel	Coherence	Null pct	RA deg	Dec deg	Null mean	Null max
raw_m1	415	0.079188	63.281%	184.643	-26.102	0.070197	0.101488
raw_m1	445	0.101265	0.781%	300.315	-26.961	0.111989	0.125530
raw_m1	480	0.103213	53.906%	249.973	-69.692	0.103087	0.113392
raw_m1	515	0.190430	95.312%	289.247	-54.535	0.184993	0.200411
raw_m1	555	0.157011	94.531%	259.247	-33.912	0.143929	0.161756
raw_m1	590	0.196796	95.312%	268.110	-52.654	0.191373	0.198842
raw_m1	630	0.140894	81.250%	275.469	-59.075	0.138316	0.154666
raw_m1	680	0.176220	97.656%	272.271	-31.015	0.161217	0.185868
raw_m1	NIR	0.213735	28.906%	263.557	-61.059	0.217954	0.232131
raw_m1	clear	0.212371	7.031%	285.195	-48.399	0.217770	0.225733
raw_m2	415	0.125528	32.812%	134.909	+19.070	0.130826	0.169404
raw_m2	445	0.073079	86.719%	108.635	+22.660	0.062255	0.090936
raw_m2	480	0.057641	24.219%	308.868	+10.605	0.063101	0.085062
raw_m2	515	0.117928	92.969%	96.728	+32.872	0.099073	0.135005
raw_m2	555	0.048937	50.000%	47.854	+29.588	0.056060	0.101160
raw_m2	590	0.116996	64.844%	329.743	-4.937	0.107932	0.129093
raw_m2	630	0.085642	58.594%	144.796	-3.753	0.085566	0.110895
raw_m2	680	0.098706	39.062%	120.591	+5.436	0.100778	0.125944
raw_m2	NIR	0.080174	49.219%	350.151	-24.845	0.076992	0.098276
raw_m2	clear	0.058056	35.938%	179.745	+1.381	0.061133	0.083624
std31_m1	415	0.050969	29.688%	193.841	-72.223	0.053848	0.070376
std31_m1	445	0.057194	61.719%	168.325	-72.265	0.057427	0.073490
std31_m1	480	0.036462	71.875%	328.890	+34.171	0.030273	0.065301
std31_m1	515	0.175980	99.219%	279.887	+16.695	0.155858	0.176514
std31_m1	555	0.050810	59.375%	134.163	+17.902	0.040491	0.073532
std31_m1	590	0.152484	64.844%	242.603	-65.813	0.150117	0.175374
std31_m1	630	0.120665	55.469%	31.369	-71.363	0.120686	0.133387
std31_m1	680	0.072204	71.875%	240.388	+67.868	0.071105	0.088112
std31_m1	NIR	0.077245	83.594%	261.804	-28.529	0.059621	0.105733
std31_m1	clear	0.075674	55.469%	66.680	-64.053	0.077716	0.105414
std31_m2	415	0.035254	42.969%	117.978	-47.611	0.032824	0.072982
std31_m2	445	0.039244	53.125%	132.527	+52.827	0.038750	0.068531
std31_m2	480	0.130837	31.250%	308.868	+15.498	0.133724	0.164127
std31_m2	515	0.059022	32.812%	158.854	+76.493	0.062750	0.079047
std31_m2	555	0.062788	46.094%	227.379	-63.694	0.063429	0.089116
std31_m2	590	0.077965	44.531%	105.718	-24.305	0.082272	0.145123
std31_m2	630	0.048446	36.719%	342.509	+12.808	0.050537	0.098313
std31_m2	680	0.041402	39.844%	337.098	-10.470	0.046969	0.081736
std31_m2	NIR	0.057878	85.156%	12.194	-47.015	0.043831	0.086038
std31_m2	clear	0.051362	0.000%	127.680	-70.496	0.084016	0.129196

Appendix D. Machine-Readable Companion and File Inventory

Role	File	SHA-256
Source record	pic_id_14C.csv	93835d3dea837ab9f7c5a43e73715e2f23c2f6874b6488db667fcd1025e1a273
Deep-analysis JSON	pic_id14C_unwrapped_cmb_deep_analysis.json	25e898be0d70c29266d3b019c351e713286902ac6f42055c55e1dfc4b3af1258
Trajectory CSV	pic_id14C_unwrapped_cmb_trajectory_5turn.csv	cf51f946a735a4feb6647db9c99d9b9cd710bc93f379f4d0833a7edec4549560
Stack-feature CSV	pic_id14C_unwrapped_cmb_stack_features.csv	f58daf40d77aeae4b60d181461a11012dac6759f30ca898586c256521d478cf
Band-relationship CSV	pic_id14C_channel_band_relationships.csv	4d373be25c93c8a03f36964527cc24a1b84902a384ef7bb3287277c359b32728
Angular-harmonic CSV	pic_id14C_angular_harmonic_relationships.csv	aabd8afb2e4e672fe92584e71f40efa4ece52d5314543a0ec9cfd17cd6ce673
Reversal-event CSV	pic_id14C_rectilinear_plane_reversal_events.csv	87e4e0b1002080a7a2989d4287edad5c179d174ea1aecfaf488907013ea3ca8b

The companion file K-Field_spectrophotometer_pic_id14C_analysis_1785888480.json contains the full source analysis object, all complete CSV tables as JSON records, the report geometry, resolved cell nodes, figure inventory, source hashes, and PDF hash. Values are stored in decimal JSON without hidden unit scaling.

Field	Value
Principal Investigator	P. Dwayne Esterline
Organization	Krytronx, LLC.
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation epoch	1785888480
Generation UTC	2026-08-05T00:08:00Z
PDF	K-Field_spectrophotometer_pic_id14C_analysis_1785888480.pdf
Machine-readable companion	K-Field_spectrophotometer_pic_id14C_analysis_1785888480.json
PDF engine	Python ReportLab
PDF fonts	Helvetica, Helvetica-Bold, Times-Roman, Courier only
Historical format references	K-Field Spectrophotometer Crossing Analysis; K-Field Spectrophotometer Optimized Geometry; K-Field 7x15 Spectrophotometer Interleaved Analysis

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